1 Algorithms (334)

Searching, Sorting, Hashing, Asymptotic worst case time and Space complexity, Algorithm design techniques: Greedy, Dynamic programming, and Divide-and-conquer, Graph search, Minimum spanning trees, Shortest paths.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximu
1 Mark Count	2	2	1	2	2	2	3	2	1	2	3
2 Marks Count	3	3	4	2	2	2	3	4	2	2.88	4
Total Marks	8	8	9	6	6	6	9	10	**6**	**7.75**	**10**

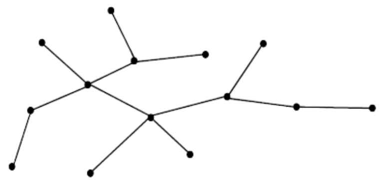
1.0.1 GATE CSE 2025 | Set 2 | Question: 49

Consider the following algorithm **someAlgo** that takes an undirected graph G as input.

someAlgo (G)

1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS.
2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS.
3. Output the distance between u and z in G.

The output of **someAlgo** (T) for the tree shown in the given figure is ___________. (Answer in integer)



gatecse2025-set2 algorithms breadth-first-search numerical-answers two-marks

Answer key☟

1.1 Algorithm Design (8)

1.1.1 Algorithm Design: GATE CSE 1992 | Question: 8

Let T be a Depth First Tree of a undirected graph G. An array P indexed by the vertices of G is given. $P[V]$ is the parent of vertex V, in T. Parent of the root is the root itself.

Give a method for finding and printing the cycle formed if the edge (u, v) of G not in T (i.e., $e \in G - T$) is now added to T.

Time taken by your method must be proportional to the length of the cycle.

Describe the algorithm in a PASCAL (C) – like language. Assume that the variables have been suitably declared.

gate1992 algorithms descriptive algorithm-design

Answer key☟

1.1.2 Algorithm Design: GATE CSE 1994 | Question: 7

An array A contains n integers in locations $A[0], A[1], \dots A[n-1]$. It is required to shift the elements of the array cyclically to the left by K places, where $1 \leq K \leq n-1$. An incomplete algorithm for doing this in linear time, without using another array is given below. Complete the algorithm by filling in the blanks. Assume all variables are suitably declared.

```
min:=n;
i=0;
while _____ do
begin
   temp:=A[i];
```

```
    j:=i;
    while ____ do
    begin
      A[j]:=____;
      j:=(j+K) mod n;
     if j<min then
      min:=j;
    end;
    A[(n+i-K)mod n]:=____;
    i:=______;
end;
```

gate1994 algorithms normal algorithm-design fill-in-the-blanks

Answer key

1.1.3 Algorithm Design: GATE CSE 2006 | Question: 17

An element in an array X is called a leader if it is greater than all elements to the right of it in X. The best algorithm to find all leaders in an array

A. solves it in linear time using a left to right pass of the array
B. solves it in linear time using a right to left pass of the array
C. solves it using divide and conquer in time $\Theta(n \log n)$
D. solves it in time $\Theta(n^2)$

gatecse-2006 algorithms normal algorithm-design

Answer key

1.1.4 Algorithm Design: GATE CSE 2006 | Question: 54

Given two arrays of numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$ where each number is 0 or 1, the fastest algorithm to find the largest span (i, j) such that $a_i + a_{i+1} + \dots + a_j = b_i + b_{i+1} + \dots + b_j$ or report that there is not such span,

A. Takes $O(3^n)$ and $\Omega(2^n)$ time if hashing is permitted
B. Takes $O(n^3)$ and $\Omega(n^{2.5})$ time in the key comparison mode
C. Takes $\Theta(n)$ time and space
D. Takes $O(\sqrt{n})$ time only if the sum of the $2n$ elements is an even number

gatecse-2006 algorithms normal algorithm-design time-complexity

Answer key

1.1.5 Algorithm Design: GATE CSE 2014 Set 1 | Question: 37

There are 5 bags labeled 1 to 5. All the coins in a given bag have the same weight. Some bags have coins of weight 10 gm, others have coins of weight 11 gm. I pick $1, 2, 4, 8, 16$ coins respectively from bags 1 to 5 Their total weight comes out to 323 gm. Then the product of the labels of the bags having 11 gm coins is ___.

gatecse-2014-set1 algorithms numerical-answers normal algorithm-design

Answer key

1.1.6 Algorithm Design: GATE CSE 2019 | Question: 25

Consider a sequence of 14 elements: $A = [-5, -10, 6, 3, -1, -2, 13, 4, -9, -1, 4, 12, -3, 0]$. The sequence sum $S(i, j) = \Sigma_{k=i}^{j} A[k]$. Determine the maximum of $S(i, j)$, where $0 \leq i \leq j < 14$. (Divide and conquer approach may be used.)

Answer: ____________

gatecse-2019 numerical-answers algorithms algorithm-design one-mark

Answer key

1.1.7 Algorithm Design: GATE CSE 2021 Set 1 | Question: 40

Define R_n to be the maximum amount earned by cutting a rod of length n meters into one or more pieces of integer length and selling them. For $i > 0$, let $p[i]$ denote the selling price of a rod whose length is i meters. Consider the array of prices:

$$\mathrm{p}[1]=1, \mathrm{p}[2]=5, \mathrm{p}[3]=8, \mathrm{p}[4]=9, \mathrm{p}[5]=10, \mathrm{p}[6]=17, \mathrm{p}[7]=18$$

Which of the following statements is/are correct about R_7?

A. $R_7 = 18$
B. $R_7 = 19$
C. R_7 is achieved by three different solutions
D. R_7 cannot be achieved by a solution consisting of three pieces

gatecse-2021-set1 multiple-selects algorithms algorithm-design two-marks

Answer key

1.1.8 Algorithm Design: GATE CSE 2024 | Set 2 | Question: 32

Consider an array X that contains n positive integers. A subarray of X is defined to be a sequence of array locations with consecutive indices.

The C code snippet given below has been written to compute the length of the longest subarray of X that contains at most two distinct integers. The code has two missing expressions labelled (P) and (Q).

```
int first=0, second=0, len1=0, len2=0, maxlen=0;
for (int i=0; i < n; i++) {
   if (X[i] == first) {
      len2++; len1++;
   } else if (X[i] == second) {
      len2++;
      len1 = ____(P)____;

      second = first;
   } else {
      len2 = ____(Q)____;

      len1 = 1; second = first;
   }
   if (len2 > maxlen) {
      maxlen = len2;
   }
   first = X[i];
}
```

Which one of the following options gives the CORRECT missing expressions?

(**Hint:** At the end of the i-th iteration, the value of $len1$ is the length of the longest subarray ending with $\mathrm{X}[i]$ that contains all equal values, and $len2$ is the length of the longest subarray ending with $\mathrm{X}[i]$ that contains at most two distinct values.)

A. $(\mathrm{P})\ \mathrm{len1}+1$ $\quad(\mathrm{Q})\ \mathrm{len2}+1$
B. $(\mathrm{P})\ 1$ $\quad(\mathrm{Q})\ \mathrm{len1}+1$
C. $(\mathrm{P})\ 1$ $\quad(\mathrm{Q})\ \mathrm{len2}+1$
D. $(\mathrm{P})\ \mathrm{len2}+1$ $\quad(\mathrm{Q})\ \mathrm{len1}+1$

gatecse2024-set2 algorithms algorithm-design two-marks

Answer key

1.2 Algorithm Design Technique (9)

1.2.1 Algorithm Design Technique: GATE CSE 1990 | Question: 12b

Consider the following problem. Given n positive integers $a_1, a_2 \dots a_n$, it is required to partition them in to two parts A and B such that, $\left|\sum_{i \in A} a_i - \sum_{i \in B} a_i\right|$ is minimised

Consider a greedy algorithm for solving this problem. The numbers are ordered so that $a_1 \geq a_2 \geq \ldots a_n$, and at i^{th} step, a_i is placed in that part whose sum in smaller at that step. Give an example with $n = 5$ for which the solution produced by the greedy algorithm is not optimal.

gate1990 descriptive algorithms algorithm-design-technique

Answer key

1.2.2 Algorithm Design Technique: GATE CSE 1990 | Question: 2-vii

Match the pairs in the following questions:

(a)	Strassen's matrix multiplication algorithm	(p)	Greedy method
(b)	Kruskal's minimum spanning tree algorithm	(q)	Dynamic programming
(c)	Biconnected components algorithm	(r)	Divide and Conquer
(d)	Floyd's shortest path algorithm	(s)	Depth-first search

gate1990 match-the-following algorithms algorithm-design-technique easy

Answer key

1.2.3 Algorithm Design Technique: GATE CSE 1994 | Question: 1.19, ISRO2016-31

Algorithm design technique used in quicksort algorithm is?

A. Dynamic programming
B. Backtracking
C. Divide and conquer
D. Greedy method

gate1994 algorithms algorithm-design-technique quick-sort easy isro2016

Answer key

1.2.4 Algorithm Design Technique: GATE CSE 1995 | Question: 1.5

Merge sort uses:

A. Divide and conquer strategy
B. Backtracking approach
C. Heuristic search
D. Greedy approach

gate1995 algorithms sorting easy algorithm-design-technique merge-sort

Answer key

1.2.5 Algorithm Design Technique: GATE CSE 1997 | Question: 1.5

The correct matching for the following pairs is

A.	All pairs shortest path	1.	Greedy
B.	Quick Sort	2.	Depth-First Search
C.	Minimum weight spanning tree	3.	Dynamic Programming
D.	Connected Components	4.	Divide and Conquer

A. A-2 B-4 C-1 D-3
B. A-3 B-4 C-1 D-2
C. A-3 B-4 C-2 D-1
D. A-4 B-1 C-2 D-3

gate1997 algorithms normal algorithm-design-technique easy match-the-following

Answer key

1.2.6 Algorithm Design Technique: GATE CSE 1998 | Question: 1.21, ISRO2008-16

Which one of the following algorithm design techniques is used in finding all pairs of shortest distances in a graph?

A. Dynamic programming
B. Backtracking
C. Greedy
D. Divide and Conquer

gate1998 algorithms algorithm-design-technique easy isro2008

Answer key

1.2.7 Algorithm Design Technique: GATE CSE 2015 Set 1 | Question: 6

Match the following:

P.	Prim's algorithm for minimum spanning tree	i.	Backtracking
Q.	Floyd-Warshall algorithm for all pairs shortest path	ii.	Greedy method
R.	Merge sort	iii.	Dynamic programming
S.	Hamiltonian circuit	iv.	Divide and conquer

A. P-iii, Q-ii, R-iv, S-i
B. P-i, Q-ii, R-iv, S-iii
C. P-ii, Q-iii, R-iv, S-i
D. P-ii, Q-i, R-iii, S-iv

gatecse-2015-set1 algorithms normal match-the-following algorithm-design-technique

Answer key

1.2.8 Algorithm Design Technique: GATE CSE 2015 Set 2 | Question: 36

Given below are some algorithms, and some algorithm design paradigms.

1.	Dijkstra's Shortest Path	i.	Divide and Conquer
2.	Floyd-Warshall algorithm to compute all pair shortest path	ii.	Dynamic Programming
3.	Binary search on a sorted array	iii.	Greedy design
4.	Backtracking search on a graph	iv.	Depth-first search
		v.	Breadth-first search

Match the above algorithms on the left to the corresponding design paradigm they follow.

A. 1-i, 2-iii, 3-i, 4-v
B. 1-iii, 2-iii, 3-i, 4-v
C. 1-iii, 2-ii, 3-i, 4-iv
D. 1-iii, 2-ii, 3-i, 4-v

gatecse-2015-set2 algorithms easy algorithm-design-technique match-the-following

Answer key

1.2.9 Algorithm Design Technique: GATE CSE 2017 Set 1 | Question: 05

Consider the following table:

	Algorithms		Design Paradigms
(P)	Kruskal	(i)	Divide and Conquer
(Q)	Quicksort	(ii)	Greedy
(R)	Floyd-Warshall	(iii)	Dynamic Programming

Match the algorithms to the design paradigms they are based on.

A. $(P) \leftrightarrow (ii), (Q) \leftrightarrow (iii), (R) \leftrightarrow (i)$
B. $(P) \leftrightarrow (iii), (Q) \leftrightarrow (i), (R) \leftrightarrow (ii)$
C. $(P) \leftrightarrow (ii), (Q) \leftrightarrow (i), (R) \leftrightarrow (iii)$
D. $(P) \leftrightarrow (i), (Q) \leftrightarrow (ii), (R) \leftrightarrow (iii)$

gatecse-2017-set1 algorithms algorithm-design-technique easy match-the-following

Answer key

1.3 Asymptotic Notation (7)

1.3.1 Asymptotic Notation: GATE CSE 1999 | Question: 2.21

If $T_1 = O(1)$, give the correct matching for the following pairs:

(M) $T_n = T_{n-1} + n$	(U) $T_n = O(n)$
(N) $T_n = T_{n/2} + n$	(V) $T_n = O(n \log n)$
(O) $T_n = T_{n/2} + n \log n$	(W) $T_n = O(n^2)$
(P) $T_n = T_{n-1} + \log n$	(X) $T_n = O(\log^2 n)$

A. M-W, N-V, O-U, P-X
B. M-W, N-U, O-X, P-V
C. M-V, N-W, O-X, P-U
D. M-W, N-U, O-V, P-X

gate1999 algorithms recurrence-relation asymptotic-notation normal match-the-following

Answer key

1.3.2 Asymptotic Notation: GATE CSE 2017 Set 1 | Question: 04

Consider the following functions from positive integers to real numbers:

$10, \sqrt{n}, n, \log_2 n, \frac{100}{n}$.
The CORRECT arrangement of the above functions in increasing order of asymptotic complexity is:

A. $\log_2 n, \frac{100}{n}, 10, \sqrt{n}, n$
B. $\frac{100}{n}, 10, \log_2 n, \sqrt{n}, n$
C. $10, \frac{100}{n}, \sqrt{n}, \log_2 n, n$
D. $\frac{100}{n}, \log_2 n, 10, \sqrt{n}, n$

gatecse-2017-set1 algorithms asymptotic-notation normal

Answer key

1.3.3 Asymptotic Notation: GATE CSE 2021 Set 1 | Question: 3

Consider the following three functions.

$$f_1 = 10^n \quad f_2 = n^{\log n} \quad f_3 = n^{\sqrt{n}}$$

Which one of the following options arranges the functions in the increasing order of asymptotic growth rate?

A. f_3, f_2, f_1
B. f_2, f_1, f_3
C. f_1, f_2, f_3
D. f_2, f_3, f_1

gatecse-2021-set1 algorithms asymptotic-notation one-mark

Answer key

1.3.4 Asymptotic Notation: GATE CSE 2022 | Question: 1

Which one of the following statements is TRUE for all positive functions $f(n)$?

A. $f(n^2) = \theta(f(n)^2)$, when $f(n)$ is a polynomial
B. $f(n^2) = o(f(n)^2)$
C. $f(n^2) = O(f(n)^2)$, when $f(n)$ is an exponential function
D. $f(n^2) = \Omega(f(n)^2)$

gatecse-2022 algorithms asymptotic-notation one-mark

Answer key

1.3.5 Asymptotic Notation: GATE CSE 2023 | Question: 19

Let f and g be functions of natural numbers given by $f(n) = n$ and $g(n) = n^2$. Which of the following statements is/are TRUE?

A. $f \in O(g)$
B. $f \in \Omega(g)$

C. $f \in o(g)$

D. $f \in \Theta(g)$

gatecse-2023 algorithms asymptotic-notation multiple-selects one-mark

Answer key

1.3.6 Asymptotic Notation: GATE CSE 2023 | Question: 44

Consider functions **Function_1** and **Function_2** expressed in pseudocode as follows:

```
Function_1
  while n>1 do
    for i=1 to n do
      x = x + 1;
    end for
    n = ⌊n/2⌋;
  end while
```

```
Function_2
  for i = 1 to 100 * n do
    x = x + 1;
  end for
```

Let $f_1(n)$ and $f_2(n)$ denote the number of times the statement "$\text{x} = \text{x} + 1$" is executed in **Function_1** and **Function_2**, respectively.

Which of the following statements is/are TRUE?

A. $f_1(n) \in \Theta(f_2(n))$

B. $f_1(n) \in o(f_2(n))$

C. $f_1(n) \in \omega(f_2(n))$

D. $f_1(n) \in O(n)$

gatecse-2023 algorithms asymptotic-notation multiple-selects two-marks

Answer key

1.3.7 Asymptotic Notation: GATE CSE 2024 | Set 2 | Question: 5

Let T(n) be the recurrence relation defined as follows:

$$T(0) = 1,$$
$$T(1) = 2, \text{ and}$$
$$T(n) = 5T(n-1) - 6T(n-2) \text{ for } n \geq 2$$

Which one of the following statements is TRUE?

A. $T(n) = \Theta(2^n)$

B. $T(n) = \Theta(n2^n)$

C. $T(n) = \Theta(3^n)$

D. $T(n) = \Theta(n3^n)$

gatecse2024-set2 algorithms recurrence-relation asymptotic-notation one-mark

Answer key

1.4 Asymptotic Notations (14)

1.4.1 Asymptotic Notations: GATE CSE 1994 | Question: 1.23

Consider the following two functions:

$$g_1(n) = \begin{cases} n^3 \text{ for } 0 \leq n \leq 10,000 \\ n^2 \text{ for } n > 10,000 \end{cases}$$

$$g_2(n) = \begin{cases} n \text{ for } 0 \leq n \leq 100 \\ n^3 \text{ for } n > 100 \end{cases}$$

Which of the following is true?

A. $g_1(n)$ is $O(g_2(n))$

B. $g_1(n)$ is $O(n^3)$

C. $g_2(n)$ is $O(g_1(n))$

D. $g_2(n)$ is $O(n)$

gate1994 algorithms asymptotic-notations normal multiple-selects

Answer key

1.4.2 Asymptotic Notations: GATE CSE 1996 | Question: 1.11

Which of the following is false?

A. $100n \log n = O(\frac{n \log n}{100})$
B. $\sqrt{\log n} = O(\log \log n)$
C. If $0 < x < y$ then $n^x = O(n^y)$
D. $2^n \neq O(nk)$

gate1996 algorithms asymptotic-notations normal

Answer key

1.4.3 Asymptotic Notations: GATE CSE 2000 | Question: 2.17

Consider the following functions

- $f(n) = 3n^{\sqrt{n}}$
- $g(n) = 2^{\sqrt{n}\log_2 n}$
- $h(n) = n!$

Which of the following is true?

A. $h(n)$ is $O(f(n))$
B. $h(n)$ is $O(g(n))$
C. $g(n)$ is not $O(f(n))$
D. $f(n)$ is $O(g(n))$

gatecse-2000 algorithms asymptotic-notations normal

Answer key

1.4.4 Asymptotic Notations: GATE CSE 2001 | Question: 1.16

Let $f(n) = n^2 \log n$ and $g(n) = n(\log n)^{10}$ be two positive functions of n. Which of the following statements is correct?

A. $f(n) = O(g(n))$ and $g(n) \neq O(f(n))$
B. $g(n) = O(f(n))$ and $f(n) \neq O(g(n))$
C. $f(n) \neq O(g(n))$ and $g(n) \neq O(f(n))$
D. $f(n) = O(g(n))$ and $g(n) = O(f(n))$

gatecse-2001 algorithms asymptotic-notations time-complexity normal

Answer key

1.4.5 Asymptotic Notations: GATE CSE 2003 | Question: 20

Consider the following three claims:

I. $(n+k)^m = \Theta(n^m)$ where k and m are constants
II. $2^{n+1} = O(2^n)$
III. $2^{2n+1} = O(2^n)$

Which of the following claims are correct?

A. I and II
B. I and III
C. II and III
D. I, II, and III

gatecse-2003 algorithms asymptotic-notations normal

Answer key

1.4.6 Asymptotic Notations: GATE CSE 2004 | Question: 29

The tightest lower bound on the number of comparisons, in the worst case, for comparison-based sorting is of the order of

A. n
B. n^2
C. $n \log n$
D. $n \log^2 n$

gatecse-2004 algorithms sorting asymptotic-notations easy

Answer key

1.4.7 Asymptotic Notations: GATE CSE 2005 | Question: 37

Suppose $T(n) = 2T(\frac{n}{2}) + n$, $T(0) = T(1) = 1$

Which one of the following is FALSE?

A. $T(n) = O(n^2)$
B. $T(n) = \Theta(n \log n)$
C. $T(n) = \Omega(n^2)$
D. $T(n) = O(n \log n)$

gatecse-2005 algorithms asymptotic-notations recurrence-relation normal

Answer key☟

1.4.8 Asymptotic Notations: GATE CSE 2008 | Question: 39

Consider the following functions:

- $f(n) = 2^n$
- $g(n) = n!$
- $h(n) = n^{\log n}$

Which of the following statements about the asymptotic behavior of $f(n)$, $g(n)$ and $h(n)$ is true?

A. $f\left(n\right)=O\left(g\left(n\right)\right); g\left(n\right)=O\left(h\left(n\right)\right)$
B. $f\left(n\right)=\Omega\left(g\left(n\right)\right); g(n)=O\left(h\left(n\right)\right)$
C. $g\left(n\right)=O\left(f\left(n\right)\right); h\left(n\right)=O\left(f\left(n\right)\right)$
D. $h\left(n\right)=O\left(f\left(n\right)\right); g\left(n\right)=\Omega\left(f\left(n\right)\right)$

gatecse-2008 algorithms asymptotic-notations normal

Answer key☟

1.4.9 Asymptotic Notations: GATE CSE 2011 | Question: 37

Which of the given options provides the increasing order of asymptotic complexity of functions f_1, f_2, f_3 and f_4?

- $f_1(n) = 2^n$
- $f_2(n) = n^{3/2}$
- $f_3(n) = n \log_2 n$
- $f_4(n) = n^{\log_2 n}$

A. f_3, f_2, f_4, f_1
B. f_3, f_2, f_1, f_4
C. f_2, f_3, f_1, f_4
D. f_2, f_3, f_4, f_1

gatecse-2011 algorithms asymptotic-notations normal

Answer key☟

1.4.10 Asymptotic Notations: GATE CSE 2012 | Question: 18

Let $W(n)$ and $A(n)$ denote respectively, the worst case and average case running time of an algorithm executed on an input of size n. Which of the following is **ALWAYS TRUE**?

A. $A(n) = \Omega(W(n))$
B. $A(n) = \Theta(W(n))$
C. $A(n) = \mathrm{O}(W(n))$
D. $A(n) = \mathrm{o}(W(n))$

gatecse-2012 algorithms easy asymptotic-notations

Answer key☟

1.4.11 Asymptotic Notations: GATE CSE 2015 Set 3 | Question: 4

Consider the equality $\displaystyle\sum_{i=0}^{n} i^3 = X$ and the following choices for X:

I. $\Theta(n^4)$
II. $\Theta(n^5)$
III. $O(n^5)$
IV. $\Omega(n^3)$

The equality above remains correct if X is replaced by

A. Only I
B. Only II
C. I or III or IV but not II
D. II or III or IV but not I

gatecse-2015-set3 algorithms asymptotic-notations normal

Answer key☟

1.4.12 Asymptotic Notations: GATE CSE 2015 Set 3 | Question: 42

Let $f(n) = n$ and $g(n) = n^{(1+\sin\ n)}$, where n is a positive integer. Which of the following statements is/are correct?

I. $f(n) = O(g(n))$
II. $f(n) = \Omega(g(n))$

A. Only I
B. Only II
C. Both I and II
D. Neither I nor II

gatecse-2015-set3 algorithms asymptotic-notations normal

Answer key☟

1.4.13 Asymptotic Notations: GATE IT 2004 | Question: 55

Let $f(n)$, $g(n)$ and $h(n)$ be functions defined for positive integers such that $f(n) = O(g(n))$, $g(n) \neq O(f(n))$, $g(n) = O(h(n))$, and $h(n) = O(g(n))$.
Which one of the following statements is FALSE?

A. $f(n) + g(n) = O(h(n) + h(n))$
B. $f(n) = O(h(n))$
C. $h(n) \neq O(f(n))$
D. $f(n)h(n) \neq O(g(n)h(n))$

gateit-2004 algorithms asymptotic-notations normal

Answer key☟

1.4.14 Asymptotic Notations: GATE IT 2008 | Question: 10

Arrange the following functions in increasing asymptotic order:

a. $n^{1/3}$
b. e^n
c. $n^{7/4}$
d. $n \log^9 n$
e. 1.0000001^n

A. a, d, c, e, b
B. d, a, c, e, b
C. a, c, d, e, b
D. a, c, d, b, e

gateit-2008 algorithms asymptotic-notations normal

Answer key☟

1.5 Bellman Ford (2)

1.5.1 Bellman Ford: GATE CSE 2009 | Question: 13

Which of the following statement(s) is/are correct regarding Bellman-Ford shortest path algorithm?

P: Always finds a negative weighted cycle, if one exists.

Q: Finds whether any negative weighted cycle is reachable from the source.

A. P only
B. Q only
C. Both P and Q
D. Neither P nor Q

gatecse-2009 algorithms graph-algorithms normal bellman-ford

Answer key☟

1.5.2 Bellman Ford: GATE CSE 2013 | Question: 19

What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

A. $\theta(n^2)$
B. $\theta(n^2 \log n)$
C. $\theta(n^3)$
D. $\theta(n^3 \log n)$

gatecse-2013 algorithms graph-algorithms normal bellman-ford

Answer key

1.6 Binary Search (3)

1.6.1 Binary Search: GATE CSE 2021 Set 2 | Question: 8

What is the worst-case number of arithmetic operations performed by recursive binary search on a sorted array of size n?

A. $\Theta(\sqrt{n})$
B. $\Theta(\log_2(n))$
C. $\Theta(n^2)$
D. $\Theta(n)$

gatecse-2021-set2 algorithms binary-search time-complexity one-mark

Answer key

1.6.2 Binary Search: GATE DA 2025 | Question: 17

For which of the following inputs does binary search take time $O(\log n)$ in the worst case?

A. An array of n integers in any order
B. A linked list of n integers in any order
C. An array of n integers in increasing order
D. A linked list of n integers in increasing order

gateda-2025 algorithms binary-search multiple-selects one-mark

Answer key

1.6.3 Binary Search: GATE DS&AI 2024 | Question: 30

Let $F(n)$ denote the maximum number of comparisons made while searching for an entry in a sorted array of size n using binary search.

Which ONE of the following options is TRUE?

A. $F(n) = F(\lfloor n/2 \rfloor) + 1$
B. $F(n) = F(\lfloor n/2 \rfloor) + F(\lceil n/2 \rceil)$
C. $F(n) = F(\lfloor n/2 \rfloor)$
D. $F(n) = F(n-1) + 1$

gate-ds-ai-2024 algorithms binary-search two-marks

Answer key

1.7 Bitonic Array (1)

1.7.1 Bitonic Array: GATE CSE 2025 | Set 2 | Question: 31

An array A of length n with distinct elements is said to be bitonic if there is an index $1 \leq i \leq n$ such that $A[1..i]$ is sorted in the non-decreasing order and $A[i+1...n]$ is sorted in the non-increasing order.

Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A ?

A. $\Theta(n)$
B. $\Theta(1)$
C. $\Theta\left(\log^2 n\right)$
D. $\Theta(\log n)$

gatecse2025-set2 algorithms searching bitonic-array time-complexity two-marks

Answer key

1.8 Depth First Search (1)

1.8.1 Depth First Search: GATE DS&AI 2024 | Question: 34

Consider a state space where the start state is number 1. The successor function for the state numbered n returns two states numbered $n+1$ and $n+2$. Assume that the states in the unexpanded state list are expanded in the ascending order of numbers and the previously expanded states are not added to the unexpanded state list.

Which ONE of the following statements about breadth-first search (BFS) and depth-first search (DFS) is true, when reaching the goal state number 6?

A. BFS expands more states than DFS.
B. DFS expands more states than BFS.
C. Both BFS and DFS expand equal number of states.
D. Both BFS and DFS do not reach the goal state number 6.

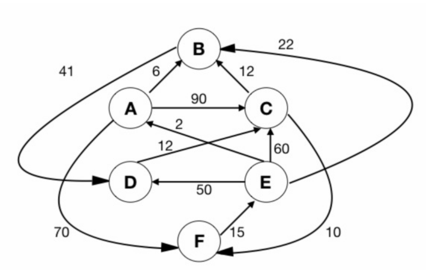
gate-ds-ai-2024 algorithms breadth-first-search depth-first-search two-marks

Answer key

1.9 Dijkstras Algorithm (6)

1.9.1 Dijkstras Algorithm: GATE CSE 1996 | Question: 17

Let G be the directed, weighted graph shown in below figure



We are interested in the shortest paths from A.

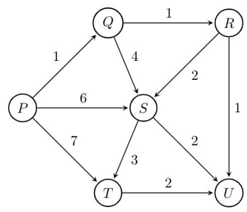
a. Output the sequence of vertices identified by the Dijkstra's algorithm for single source shortest path when the algorithm is started at node A
b. Write down sequence of vertices in the shortest path from A to E
c. What is the cost of the shortest path from A to E?

gate1996 algorithms graph-algorithms normal dijkstras-algorithm descriptive

Answer key

1.9.2 Dijkstras Algorithm: GATE CSE 2004 | Question: 44

Suppose we run Dijkstra's single source shortest path algorithm on the following edge-weighted directed graph with vertex P as the source.



In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized?

A. P,Q,R,S,T,U B. P,Q,R,U,S,T C. P,Q,R,U,T,S D. P,Q,T,R,U,S

gatecse-2004 algorithms graph-algorithms normal dijkstras-algorithm

Answer key

1.9.3 Dijkstras Algorithm: GATE CSE 2005 | Question: 38

Let $G(V, E)$ be an undirected graph with positive edge weights. Dijkstra's single source shortest path algorithm can be implemented using the binary heap data structure with time complexity:

A. $O\left(|V|^2\right)$
B. $O(|E| + |V| \log |V|)$
C. $O(|V| \log |V|)$
D. $O((|E| + |V|) \log |V|)$

gatecse-2005 algorithms graph-algorithms normal dijkstras-algorithm

Answer key

1.9.4 Dijkstras Algorithm: GATE CSE 2006 | Question: 12

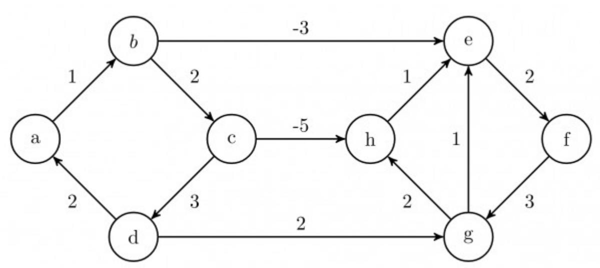
To implement Dijkstra's shortest path algorithm on unweighted graphs so that it runs in linear time, the data structure to be used is:

A. Queue
B. Stack
C. Heap
D. B-Tree

gatecse-2006 algorithms graph-algorithms easy dijkstras-algorithm

Answer key

1.9.5 Dijkstras Algorithm: GATE CSE 2008 | Question: 45



Dijkstra's single source shortest path algorithm when run from vertex a in the above graph, computes the correct shortest path distance to

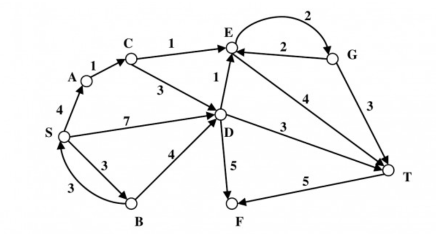
A. only vertex a
B. only vertices a, e, f, g, h
C. only vertices a, b, c, d
D. all the vertices

gatecse-2008 algorithms graph-algorithms normal dijkstras-algorithm

Answer key

1.9.6 Dijkstras Algorithm: GATE CSE 2012 | Question: 40

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



A. SDT
B. SBDT
C. SACDT
D. SACET

gatecse-2012 algorithms graph-algorithms normal dijkstras-algorithm

Answer key

1.10 Directed Graph (1)

1.10.1 Directed Graph: GATE DA 2025 | Question: 55

Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \ldots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2,$ for all $i, j \in V\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is ____________ (Answer in integer)

gateda-2025 algorithms depth-first-search directed-graph numerical-answers two-marks

1.11 Double Hashing (1)

1.11.1 Double Hashing: GATE CSE 2025 | Set 1 | Question: 55

In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i-th probe in the open address table is $[h_1(k) + ih_2(k)] \bmod m$. The following keys are inserted in the given order: $63, 50, 25, 79, 67, 24$.

The slot at which key 24 gets stored is ___________ . (Answer in integer)

gatecse2025-set1 algorithms hashing double-hashing numerical-answers easy two-marks

Answer key

1.12 Dynamic Programming (9)

1.12.1 Dynamic Programming: GATE CSE 2008 | Question: 80

The subset-sum problem is defined as follows. Given a set of n positive integers, $S = \{a_1, a_2, a_3, \ldots, a_n\}$, and positive integer W, is there a subset of S whose elements sum to W? A dynamic program for solving this problem uses a 2-dimensional Boolean array, X, with n rows and $W + 1$ columns. $X[i, j], 1 \leq i \leq n, 0 \leq j \leq W$, is TRUE, if and only if there is a subset of $\{a_1, a_2, \ldots, a_i\}$ whose elements sum to j.

Which of the following is valid for $2 \leq i \leq n$, and $a_i \leq j \leq W$?

A. $X[i, j] = X[i - 1, j] \vee X[i, j - a_i]$
B. $X[i, j] = X[i - 1, j] \vee X[i - 1, j - a_i]$
C. $X[i, j] = X[i - 1, j] \wedge X[i, j - a_i]$
D. $X[i, j] = X[i - 1, j] \wedge X[i - 1, j - a_i]$

gatecse-2008 algorithms normal dynamic-programming

Answer key

1.12.2 Dynamic Programming: GATE CSE 2008 | Question: 81

The subset-sum problem is defined as follows. Given a set of n positive integers, $S = \{a_1, a_2, a_3, \ldots, a_n\}$, and positive integer W, is there a subset of S whose elements sum to W? A dynamic program for solving this problem uses a 2-dimensional Boolean array, X, with n rows and $W + 1$ columns. $X[i, j], 1 \leq i \leq n, 0 \leq j \leq W$, is TRUE, if and only if there is a subset of $\{a_1, a_2, \ldots, a_i\}$ whose elements sum to j.

Which entry of the array X, if TRUE, implies that there is a subset whose elements sum to W?

A. $X[1, W]$ B. $X[n, 0]$ C. $X[n, W]$ D. $X[n - 1, n]$

gatecse-2008 algorithms normal dynamic-programming

Answer key

1.12.3 Dynamic Programming: GATE CSE 2009 | Question: 53

A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences $X[m]$ and $Y[n]$ of lengths m and n, respectively with indexes of X and Y starting from 0.

We wish to find the length of the longest common sub-sequence (LCS) of $X[m]$ and $Y[n]$ as $l(m, n)$, where an incomplete recursive definition for the function $I(i,j)$ to compute the length of the LCS of $X[m]$ and $Y[n]$ is given below:

I(i,j) = 0, if either i = 0 or j = 0
= expr1, if i,j > 0 and X[i-1] = Y[j-1]
= expr2, if i,j > 0 and X[i-1] ≠ Y[j-1]

Which one of the following options is correct?

A. $\text{expr1} = l\,(i-1,j)+1$
B. $\text{expr1} = l\,(i,j-1)$
C. $\text{expr2} = \max\left(l\,(i-1,j),l\,(i,j-1)\right)$
D. $\text{expr2} = \max\left(l\,(i-1,j-1),l\,(i,j)\right)$

gatecse-2009 algorithms normal dynamic-programming recursion

Answer key

1.12.4 Dynamic Programming: GATE CSE 2009 | Question: 54

A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences $X[m]$ and $Y[n]$ of lengths m and n, respectively with indexes of X and Y starting from 0.

We wish to find the length of the longest common sub-sequence (LCS) of $X[m]$ and $Y[n]$ as $l(m,n)$, where an incomplete recursive definition for the function $I(i,j)$ to compute the length of the LCS of $X[m]$ and $Y[n]$ is given below:

I(i,j) = 0, if either i = 0 or j = 0
= expr1, if i,j > 0 and X[i-1] = Y[j-1]
= expr2, if i,j > 0 and X[i-1] ≠ Y[j-1]

The value of $l(i,j)$ could be obtained by dynamic programming based on the correct recursive definition of $l(i,j)$ of the form given above, using an array $L[M,N]$, where $M=m+1$ and $N=n+1$, such that $L[i,j]=l(i,j)$.

Which one of the following statements would be TRUE regarding the dynamic programming solution for the recursive definition of $l(i,j)$?

A. All elements of L should be initialized to 0 for the values of $l(i,j)$ to be properly computed.
B. The values of $l(i,j)$ may be computed in a row major order or column major order of $L[M,N]$.
C. The values of $l(i,j)$ cannot be computed in either row major order or column major order of $L[M,N]$.
D. $L[p,q]$ needs to be computed before $L[r,s]$ if either $p<r$ or $q<s$.

gatecse-2009 normal algorithms dynamic-programming recursion

Answer key

1.12.5 Dynamic Programming: GATE CSE 2010 | Question: 34

The weight of a sequence $a_0,a_1,\dots,a_{n-1}$ of real numbers is defined as $a_0+a_1/2+\dots+a_{n-1}/2^{n-1}$. A subsequence of a sequence is obtained by deleting some elements from the sequence, keeping the order of the remaining elements the same. Let X denote the maximum possible weight of a subsequence of $a_o,a_1,\dots,a_{n-1}$ and Y the maximum possible weight of a subsequence of $a_1,a_2,\dots,a_{n-1}$. Then X is equal to

A. $max(Y,a_0+Y)$
B. $max(Y,a_0+Y/2)$
C. $max(Y,a_0+2Y)$
D. $a_0+Y/2$

gatecse-2010 algorithms dynamic-programming normal

Answer key

1.12.6 Dynamic Programming: GATE CSE 2011 | Question: 25

An algorithm to find the length of the longest monotonically increasing sequence of numbers in an array $A[0:n-1]$ is given below.

Let L_i, denote the length of the longest monotonically increasing sequence starting at index i in the array.

Initialize $L_{n-1}=1$.

For all i such that $0\leq i\leq n-2$

$$L_i = \begin{cases} 1 + L_{i+1} & \text{if A[i] < A[i+1]} \\ 1 & \text{Otherwise} \end{cases}$$

Finally, the length of the longest monotonically increasing sequence is $\max\left(L_0, L_1, \ldots, L_{n-1}\right)$.

Which of the following statements is **TRUE**?

A. The algorithm uses dynamic programming paradigm
B. The algorithm has a linear complexity and uses branch and bound paradigm
C. The algorithm has a non-linear polynomial complexity and uses branch and bound paradigm
D. The algorithm uses divide and conquer paradigm

gatecse-2011 algorithms easy dynamic-programming

Answer key

1.12.7 Dynamic Programming: GATE CSE 2014 Set 2 | Question: 37

Consider two strings A = *"qpqrr"* and B = *"pqprqrp"*. Let x be the length of the longest common subsequence *(not necessarily contiguous)* between A and B and let y be the number of such longest common subsequences between A and B. Then $x + 10y =$___.

gatecse-2014-set2 algorithms normal numerical-answers dynamic-programming

Answer key

1.12.8 Dynamic Programming: GATE CSE 2014 Set 3 | Question: 37

Suppose you want to move from 0 to 100 on the number line. In each step, you either move right by a unit distance or you take a *shortcut*. A shortcut is simply a pre-specified pair of integers $i,\ j$ with $i < j$. Given a shortcut (i, j), if you are at position i on the number line, you may directly move to j. Suppose $T(k)$ denotes the smallest number of steps needed to move from k to 100. Suppose further that there is at most 1 shortcut involving any number, and in particular, from 9 there is a shortcut to 15. Let y and z be such that $T(9) = 1 + \min(T(y), T(z))$. Then the value of the product yz is _____.

gatecse-2014-set3 algorithms normal numerical-answers dynamic-programming

Answer key

1.12.9 Dynamic Programming: GATE CSE 2016 Set 2 | Question: 14

The Floyd-Warshall algorithm for all-pair shortest paths computation is based on

A. Greedy paradigm.
B. Divide-and-conquer paradigm.
C. Dynamic Programming paradigm.
D. Neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.

gatecse-2016-set2 algorithms dynamic-programming easy

Answer key

1.13 Graph Algorithms (11)

1.13.1 Graph Algorithms: GATE CSE 1994 | Question: 1.22

Which of the following statements is false?

A. Optimal binary search tree construction can be performed efficiently using dynamic programming
B. Breadth-first search cannot be used to find connected components of a graph
C. Given the prefix and postfix walks over a binary tree, the binary tree cannot be uniquely constructed.
D. Depth-first search can be used to find connected components of a graph

gate1994 algorithms normal graph-algorithms

Answer key

1.13.2 Graph Algorithms: GATE CSE 2003 | Question: 70

Let $G = (V, E)$ be a directed graph with n vertices. A path from v_i to v_j in G is a sequence of vertices ($v_i, v_{i+1}, \ldots, v_j$) such that $(v_k, v_{k+1}) \in E$ for all k in i through $j-1$. A simple path is a path in which no vertex appears more than once.

Let A be an $n \times n$ array initialized as follows:

$$A[j, k] = \begin{cases} 1, & \text{if } (j, k) \in E \\ 0, & \text{otherwise} \end{cases}$$

Consider the following algorithm:

```
for i=1 to n
   for j=1 to n
      for k=1 to n
         A[j,k] = max(A[j,k], A[j,i] + A[i,k]);
```

Which of the following statements is necessarily true for all j and k after termination of the above algorithm?

A. $A[j, k] \leq n$
B. If $A[j, j] \geq n-1$ then G has a Hamiltonian cycle
C. If there exists a path from j to k, $A[j, k]$ contains the longest path length from j to k
D. If there exists a path from j to k, every simple path from j to k contains at most $A[j, k]$ edges

gatecse-2003 algorithms graph-algorithms normal

Answer key

1.13.3 Graph Algorithms: GATE CSE 2005 | Question: 82a

Let s and t be two vertices in a undirected graph $G = (V, E)$ having distinct positive edge weights. Let $[X, Y]$ be a partition of V such that $s \in X$ and $t \in Y$. Consider the edge e having the minimum weight amongst all those edges that have one vertex in X and one vertex in Y.

The edge e must definitely belong to:

A. the minimum weighted spanning tree of G
B. the weighted shortest path from s to t
C. each path from s to t
D. the weighted longest path from s to t

gatecse-2005 algorithms graph-algorithms normal

Answer key

1.13.4 Graph Algorithms: GATE CSE 2005 | Question: 82b

Let s and t be two vertices in a undirected graph $G = (V, E)$ having distinct positive edge weights. Let $[X, Y]$ be a partition of V such that $s \in X$ and $t \in Y$. Consider the edge e having the minimum weight amongst all those edges that have one vertex in X and one vertex in Y.

Let the weight of an edge e denote the congestion on that edge. The congestion on a path is defined to be the maximum of the congestions on the edges of the path. We wish to find the path from s to t having minimum congestion. Which of the following paths is always such a path of minimum congestion?

A. a path from s to t in the minimum weighted spanning tree
B. a weighted shortest path from s to t
C. an Euler walk from s to t
D. a Hamiltonian path from s to t

gatecse-2005 algorithms graph-algorithms normal

Answer key

1.13.5 Graph Algorithms: GATE CSE 2016 Set 2 | Question: 41

In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u, and $[u]$ in the adjacency list of v. These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E| = m$ and $|V| = n$, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

A. $\Theta\left(n^2\right)$
B. $\Theta\left(n+m\right)$
C. $\Theta\left(m^2\right)$
D. $\Theta\left(n^4\right)$

gatecse-2016-set2 algorithms graph-algorithms normal

Answer key☟

1.13.6 Graph Algorithms: GATE CSE 2017 Set 1 | Question: 26

Let $G=(V,E)$ be *any* connected, undirected, edge-weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements:

I. Minimum Spanning Tree of G is always unique.
II. Shortest path between any two vertices of G is always unique.

Which of the above statements is/are necessarily true?

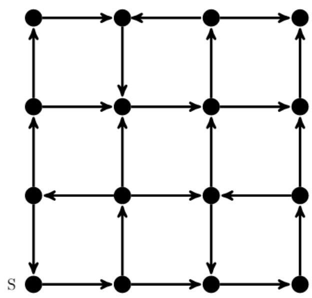
A. I only
B. II only
C. both I and II
D. neither I nor II

gatecse-2017-set1 algorithms graph-algorithms normal

Answer key☟

1.13.7 Graph Algorithms: GATE CSE 2021 Set 2 | Question: 46

Consider the following directed graph:



Which of the following is/are correct about the graph?

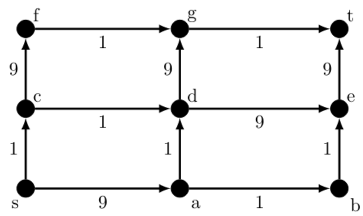
A. The graph does not have a topological order
B. A depth-first traversal starting at vertex S classifies three directed edges as back edges
C. The graph does not have a strongly connected component
D. For each pair of vertices u and v, there is a directed path from u to v

gatecse-2021-set2 multiple-selects algorithms graph-algorithms two-marks

Answer key☟

1.13.8 Graph Algorithms: GATE CSE 2021 Set 2 | Question: 55

In a directed acyclic graph with a source vertex **s**, the *quality-score* of a directed path is defined to be the product of the weights of the edges on the path. Further, for a vertex v other than **s**, the quality-score of v is defined to be the maximum among the quality-scores of all the paths from **s** to v. The quality-score of **s** is assumed to be 1.



The sum of the quality-scores of all vertices on the graph shown above is _______ gatecse-2021-set2 algorithms graph-algorithms directed-acyclic-graph numerical-answers two-marks

Answer key

1.13.9 Graph Algorithms: GATE IT 2005 | Question: 15

In the following table, the left column contains the names of standard graph algorithms and the right column contains the time complexities of the algorithms. Match each algorithm with its time complexity.

1.	Bellman-Ford algorithm	A:	$O(m \log n)$
2.	Kruskal's algorithm	B:	$O(n^3)$
3.	Floyd-Warshall algorithm	C:	$O(nm)$
4.	Topological sorting	D:	$O(n+m)$

A. $1 \rightarrow \text{C}, 2 \rightarrow \text{A}, 3 \rightarrow \text{B}, 4 \rightarrow \text{D}$

B. $1 \rightarrow \text{B}, 2 \rightarrow \text{D}, 3 \rightarrow \text{C}, 4 \rightarrow \text{A}$

C. $1 \rightarrow \text{C}, 2 \rightarrow \text{D}, 3 \rightarrow \text{A}, 4 \rightarrow \text{B}$

D. $1 \rightarrow \text{B}, 2 \rightarrow \text{A}, 3 \rightarrow \text{C}, 4 \rightarrow \text{D}$

gateit-2005 algorithms graph-algorithms match-the-following easy

Answer key

1.13.10 Graph Algorithms: GATE IT 2005 | Question: 84a

A sink in a directed graph is a vertex i such that there is an edge from every vertex $j \neq i$ to i and there is no edge from i to any other vertex. A directed graph G with n vertices is represented by its adjacency matrix A, where $A[i][j] = 1$ if there is an edge directed from vertex i to j and 0 otherwise. The following algorithm determines whether there is a sink in the graph G.

```
i = 0;
do {
   j = i + 1;
   while ((j < n) && E1) j++;
   if (j < n) E2;
} while (j < n);
flag = 1;
for (j = 0; j < n; j++)
   if ((j! = i) && E3) flag = 0;
if (flag) printf("Sink exists");
else printf ("Sink does not exist");
```

Choose the correct expressions for E_1 and E_2

A. $E_1 : A[i][j]$ and $E_2 : i = j;$

B. $E_1 :\ !A[i][j]$ and $E_2 : i = j + 1;$

C. $E_1 :\ !A[i][j]$ and $E_2 : i = j;$

D. $E_1 : A[i][j]$ and $E_2 : i = j + 1;$

gateit-2005 algorithms graph-algorithms normal

Answer key

1.13.11 Graph Algorithms: GATE IT 2005 | Question: 84b

A sink in a directed graph is a vertex i such that there is an edge from every vertex $j \neq i$ to i and there is no edge from i to any other vertex. A directed graph G with n vertices is represented by its adjacency matrix A, where $A[i][j] = 1$ if there is an edge directed from vertex i to j and 0 otherwise. The following algorithm determines whether there is a sink in the graph G.

```
i = 0;
do {
   j = i + 1;
   while ((j < n) && E1) j++;
   if (j < n) E2;
} while (j < n);
flag = 1;
for (j = 0; j < n; j++)
   if ((j! = i) && E3) flag = 0;
if (flag) printf("Sink exists") ;
else printf ("Sink does not exist");
```

Choose the correct expression for E_3

A. $(A[i][j]\ \&\&\ !A[j][i])$

B. $(!A[i][j]\ \&\&\ A[j][i])$

C. $(!A[i][j] \ || \ A[j][i])$

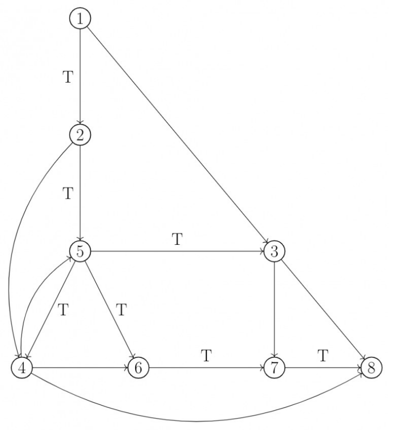
D. $(A[i][j] \ || \ !A[j][i])$

gateit-2005 algorithms graph-algorithms normal

Answer key

1.14 Graph Search (22)

1.14.1 Graph Search: GATE CSE 1989 | Question: 4-vii



In the graph shown above, the depth-first spanning tree edges are marked with a $'T'$. Identify the forward, backward, and cross edges.

gate1989 descriptive algorithms graph-algorithms depth-first-search graph-search

Answer key

1.14.2 Graph Search: GATE CSE 2000 | Question: 1.13

The most appropriate matching for the following pairs

X: depth first search	1: heap
Y: breadth first search	2: queue
Z: sorting	3: stack

is:

A. $\text{X - 1, Y - 2, Z - 3}$

B. $\text{X - 3, Y - 1, Z - 2}$

C. $\text{X - 3, Y - 2, Z - 1}$

D. $\text{X - 2, Y - 3, Z - 1}$

gatecse-2000 algorithms easy graph-algorithms graph-search match-the-following

Answer key

1.14.3 Graph Search: GATE CSE 2000 | Question: 2.19

Let G be an undirected graph. Consider a depth-first traversal of G, and let T be the resulting depth-first search tree. Let u be a vertex in G and let v be the first new (unvisited) vertex visited after visiting u in the traversal. Which of the following statement is always true?

A. $\{u, v\}$ must be an edge in G, and u is a descendant of v in T
B. $\{u, v\}$ must be an edge in G, and v is a descendant of u in T
C. If $\{u, v\}$ is not an edge in G then u is a leaf in T
D. If $\{u, v\}$ is not an edge in G then u and v must have the same parent in T

gatecse-2000 algorithms graph-algorithms normal graph-search

Answer key

1.14.4 Graph Search: GATE CSE 2001 | Question: 2.14

Consider an undirected, unweighted graph G. Let a breadth-first traversal of G be done starting from a node r. Let $d(r,u)$ and $d(r,v)$ be the lengths of the shortest paths from r to u and v respectively in G. If u is visited before v during the breadth-first traversal, which of the following statements is correct?

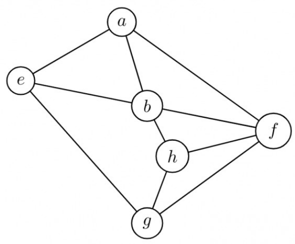
A. $d(r,u) < d(r,v)$
B. $d(r,u) > d(r,v)$
C. $d(r,u) \leq d(r,v)$
D. None of the above

gatecse-2001 algorithms graph-algorithms normal graph-search

Answer key☟

1.14.5 Graph Search: GATE CSE 2003 | Question: 21

Consider the following graph:



Among the following sequences:

I. abeghf
II. abfehg
III. abfhge
IV. afghbe

Which are the depth-first traversals of the above graph?

A. I, II and IV only
B. I and IV only
C. II, III and IV only
D. I, III and IV only

gatecse-2003 algorithms graph-algorithms normal graph-search

Answer key☟

1.14.6 Graph Search: GATE CSE 2006 | Question: 48

Let T be a depth first search tree in an undirected graph G. Vertices u and ν are leaves of this tree T. The degrees of both u and ν in G are at least 2. which one of the following statements is true?

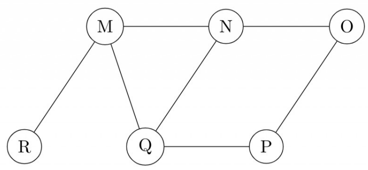
A. There must exist a vertex w adjacent to both u and ν in G
B. There must exist a vertex w whose removal disconnects u and ν in G
C. There must exist a cycle in G containing u and ν
D. There must exist a cycle in G containing u and all its neighbours in G

gatecse-2006 algorithms graph-algorithms normal graph-search

Answer key☟

1.14.7 Graph Search: GATE CSE 2008 | Question: 19

The Breadth First Search algorithm has been implemented using the queue data structure. One possible order of visiting the nodes of the following graph is:



A. MNOPQR
B. NQMPOR
C. QMNPRO
D. QMNPOR

gatecse-2008 normal algorithms graph-algorithms graph-search

Answer key

1.14.8 Graph Search: GATE CSE 2014 Set 1 | Question: 11

Let G be a graph with n vertices and m edges. What is the tightest upper bound on the running time of Depth First Search on G, when G is represented as an adjacency matrix?

A. $\Theta(n)$ B. $\Theta(n+m)$ C. $\Theta(n^2)$ D. $\Theta(m^2)$

gatecse-2014-set1 algorithms graph-algorithms normal graph-search

Answer key

1.14.9 Graph Search: GATE CSE 2014 Set 2 | Question: 14

Consider the tree arcs of a BFS traversal from a source node W in an unweighted, connected, undirected graph. The tree T formed by the tree arcs is a data structure for computing

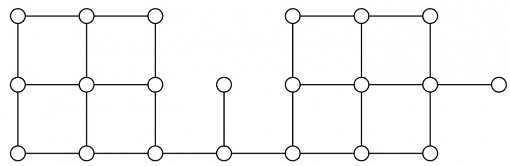
A. the shortest path between every pair of vertices.
B. the shortest path from W to every vertex in the graph.
C. the shortest paths from W to only those nodes that are leaves of T.
D. the longest path in the graph.

gatecse-2014-set2 algorithms graph-algorithms normal graph-search

Answer key

1.14.10 Graph Search: GATE CSE 2014 Set 3 | Question: 13

Suppose depth first search is executed on the graph below starting at some unknown vertex. Assume that a recursive call to visit a vertex is made only after first checking that the vertex has not been visited earlier. Then the maximum possible recursion depth (including the initial call) is _________.



gatecse-2014-set3 algorithms graph-algorithms numerical-answers normal graph-search

Answer key

1.14.11 Graph Search: GATE CSE 2015 Set 1 | Question: 45

Let $G=(V,E)$ be a simple undirected graph, and s be a particular vertex in it called the source. For $x \in V$, let $d(x)$ denote the shortest distance in G from s to x. A breadth first search (BFS) is performed starting at s. Let T be the resultant BFS tree. If (u,v) is an edge of G that is not in T, then which one of the following CANNOT be the value of $d(u)-d(v)$?

A. -1 B. 0 C. 1 D. 2

gatecse-2015-set1 algorithms graph-algorithms normal graph-search

Answer key

1.14.12 Graph Search: GATE CSE 2016 Set 2 | Question: 11

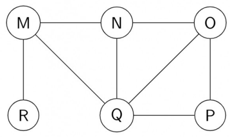
Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n^{th} vertex in this BFS traversal, then the maximum possible value of n is __________

gatecse-2016-set2 algorithms graph-algorithms normal numerical-answers graph-search

Answer key

1.14.13 Graph Search: GATE CSE 2017 Set 2 | Question: 15

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



A. MNOPQR B. NQMPOR C. QMNROP D. POQNMR

gatecse-2017-set2 algorithms graph-algorithms graph-search

Answer key

1.14.14 Graph Search: GATE CSE 2018 | Question: 30

Let G be a simple undirected graph. Let T_D be a depth first search tree of G. Let T_B be a breadth first search tree of G. Consider the following statements.

I. No edge of G is a cross edge with respect to T_D. (A cross edge in G is between two nodes neither of which is an ancestor of the other in T_D).
II. For every edge (u, v) of G, if u is at depth i and v is at depth j in T_B, then $|i - j| = 1$.

Which of the statements above must necessarily be true?

A. I only B. II only C. Both I and II D. Neither I nor II

gatecse-2018 algorithms graph-algorithms graph-search normal two-marks

Answer key

1.14.15 Graph Search: GATE CSE 2023 | Question: 46

Let $U = \{1, 2, 3\}$. Let 2^U denote the powerset of U. Consider an undirected graph G whose vertex set is 2^U. For any $A, B \in 2^U$, (A, B) is an edge in G if and only if (i) $A \neq B$, and (ii) either $A \subsetneq B$ or $B \subsetneq A$. For any vertex A in G, the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by $\mathcal{B}(A)$.

If $\emptyset$ denotes the empty set, then the cardinality of $\mathcal{B}(\emptyset)$ is ______________.

gatecse-2023 algorithms breadth-first-search numerical-answers two-marks graph-search

Answer key

1.14.16 Graph Search: GATE CSE 2024 | Set 1 | Question: 35

Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v. Suppose T is also a breadth first search (BFS) tree in G, rooted at v. Which of the following statements is/are TRUE for *every* such graph G and tree T?

A. There are no back-edges in G with respect to the tree T
B. There are no cross-edges in G with respect to the tree T
C. There are no forward-edge in G with respect to the tree T
D. The only edges in G are the edges in T

gatecse2024-set1 algorithms multiple-selects graph-search two-marks

Answer key

1.14.17 Graph Search: GATE CSE 2024 | Set 1 | Question: 50

The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is __________.

gatecse2024-set1 numerical-answers graph-search graph-algorithms two-marks

Answer key☟

1.14.18 Graph Search: GATE DS&AI 2024 | Question: 4

Consider performing depth-first search (DFS) on an undirected and unweighted graph G starting at vertex s. For any vertex u in $G, d[u]$ is the length of the shortest path from s to u. Let (u, v) be an edge in G such that $d[u] < d[v]$. If the edge (u, v) is explored first in the direction from u to v during the above DFS, then (u, v) becomes a ________ edge.

A. tree
B. cross
C. back
D. gray

gate-ds-ai-2024 graph-search depth-first-search algorithms one-mark

Answer key☟

1.14.19 Graph Search: GATE IT 2005 | Question: 14

In a depth-first traversal of a graph G with n vertices, k edges are marked as tree edges. The number of connected components in G is

A. k
B. $k+1$
C. $n-k-1$
D. $n-k$

gateit-2005 algorithms graph-algorithms normal graph-search

Answer key☟

1.14.20 Graph Search: GATE IT 2006 | Question: 47

Consider the depth-first-search of an undirected graph with 3 vertices P, Q, and R. Let discovery time $d(u)$ represent the time instant when the vertex u is first visited, and finish time $f(u)$ represent the time instant when the vertex u is last visited. Given that

$d(P) = 5$ units	$f(P) = 12$ units
$d(Q) = 6$ units	$f(Q) = 10$ units
$d(R) = 14$ unit	$f(R) = 18$ units

Which one of the following statements is TRUE about the graph?

A. There is only one connected component
B. There are two connected components, and P and R are connected
C. There are two connected components, and Q and R are connected
D. There are two connected components, and P and Q are connected

gateit-2006 algorithms graph-algorithms normal graph-search depth-first-search

Answer key☟

1.14.21 Graph Search: GATE IT 2007 | Question: 24

A depth-first search is performed on a directed acyclic graph. Let $d[u]$ denote the time at which vertex u is visited for the first time and $f[u]$ the time at which the DFS call to the vertex u terminates. Which of the following statements is always TRUE for all edges (u, v) in the graph ?

A. $d[u] < d[v]$
B. $d[u] < f[v]$
C. $f[u] < f[v]$
D. $f[u] > f[v]$

gateit-2007 algorithms graph-algorithms normal graph-search depth-first-search

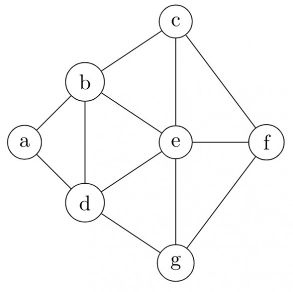
Answer key☟

1.14.22 Graph Search: GATE IT 2008 | Question: 47

Consider the following sequence of nodes for the undirected graph given below:

1. $a\,b\,e\,f\,d\,g\,c$
2. $a\,b\,e\,f\,c\,g\,d$
3. $a\,d\,g\,e\,b\,c\,f$
4. $a\,d\,b\,c\,g\,e\,f$

A Depth First Search (DFS) is started at node a. The nodes are listed in the order they are first visited. Which of the above is/are possible output(s)?



A. 1 and 3 only B. 2 and 3 only C. $2,3$ and 4 only D. $1,2$ and 3 only

gateit-2008 algorithms graph-algorithms normal graph-search depth-first-search

Answer key☟

1.15 Greedy Algorithms (5)

1.15.1 Greedy Algorithms: GATE CSE 1999 | Question: 2.20

The minimum number of record movements required to merge five files A (with 10 records), B (with 20 records), C (with 15 records), D (with 5 records) and E (with 25 records) is:

A. 165 B. 90 C. 75 D. 65

gate1999 algorithms normal greedy-algorithms

Answer key☟

1.15.2 Greedy Algorithms: GATE CSE 2003 | Question: 69

The following are the starting and ending times of activities A, B, C, D, E, F, G and H respectively in chronological order: "$a_s\, b_s\, c_s\, a_e\, d_s\, c_e\, e_s\, f_s\, b_e\, d_e\, g_s\, e_e\, f_e\, h_s\, g_e\, h_e$". Here, x_s denotes the starting time and x_e denotes the ending time of activity X. We need to schedule the activities in a set of rooms available to us. An activity can be scheduled in a room only if the room is reserved for the activity for its entire duration. What is the minimum number of rooms required?

A. 3 B. 4 C. 5 D. 6

gatecse-2003 algorithms normal greedy-algorithms

Answer key☟

1.15.3 Greedy Algorithms: GATE CSE 2005 | Question: 84a

We are given 9 tasks $T_1, T_2, \dots, T_9$. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i. Profit P_i is earned if the task is completed before the end of the d_i^{th} unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

Are all tasks completed in the schedule that gives maximum profit?

A. All tasks are completed B. T_1 and T_6 are left out
C. T_1 and T_8 are left out D. T_4 and T_6 are left out

gatecse-2005 algorithms greedy-algorithms process-scheduling normal

Answer key

1.15.4 Greedy Algorithms: GATE CSE 2005 | Question: 84b

We are given 9 tasks $T_1, T_2, \dots, T_9$. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i. Profit P_i is earned if the task is completed before the end of the d_i^{th} unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

What is the maximum profit earned?

A. 147
B. 165
C. 167
D. 175

gatecse-2005 algorithms greedy-algorithms process-scheduling normal

Answer key

1.15.5 Greedy Algorithms: GATE CSE 2018 | Question: 48

Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	**Weight (in Kgs)**	**Value (in rupees)**
1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy}.

The value of $V_{opt} - V_{greedy}$ is _____

gatecse-2018 algorithms greedy-algorithms numerical-answers two-marks

Answer key

1.16 Hashing (7)

1.16.1 Hashing: GATE CSE 1989 | Question: 1-vii, ISRO2015-14

A hash table with ten buckets with one slot per bucket is shown in the following figure. The symbols $S1$ to $S7$ initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

0	S7
1	S1
2	
3	S4
4	S2
5	
6	S5
7	
8	S6
9	S3

A. 4 B. 5 C. 6 D. 3

hashing isro2015 gate1989 algorithms normal

Answer key

1.16.2 Hashing: GATE CSE 1990 | Question: 13b

Consider a hash table with chaining scheme for overflow handling:

i. What is the worst-case timing complexity of inserting n elements into such a table?
ii. For what type of instance does this hashing scheme take the worst-case time for insertion?

gate1990 hashing algorithms descriptive

Answer key

1.16.3 Hashing: GATE CSE 2020 | Question: 23

Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 90$ is____________.

gatecse-2020 numerical-answers algorithms hashing one-mark

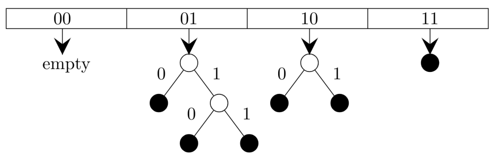
Answer key

1.16.4 Hashing: GATE CSE 2021 Set 1 | Question: 47

Consider a *dynamic* hashing approach for 4-bit integer keys:

1. There is a main hash table of size 4.
2. The 2 least significant bits of a key is used to index into the main hash table.
3. Initially, the main hash table entries are empty.
4. Thereafter, when more keys are hashed into it, to resolve collisions, the set of all keys corresponding to a main hash table entry is organized as a binary tree that grows on demand.
5. First, the 3^{rd} least significant bit is used to divide the keys into left and right subtrees.
6. To resolve more collisions, each node of the binary tree is further sub-divided into left and right subtrees based on the 4^{th} least significant bit.
7. A split is done only if it is needed, i.e., only when there is a collision.

Consider the following state of the hash table.



Which of the following sequences of key insertions can cause the above state of the hash table (assume the keys are in decimal notation)?

A. $5, 9, 4, 13, 10, 7$
B. $9, 5, 10, 6, 7, 1$
C. $10, 9, 6, 7, 5, 13$
D. $9, 5, 13, 6, 10, 14$

gatecse-2021-set1 multiple-selects algorithms hashing two-marks

Answer key

1.16.5 Hashing: GATE CSE 2022 | Question: 6

Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h_1 and h_2. Further suppose our hashing scheme uses h_1 for the odd keys and h_2 for the even keys. What is the expected number of keys in a slot?

A. $\frac{m}{n}$ B. $\frac{n}{m}$ C. $\frac{2n}{m}$ D. $\frac{n}{2m}$

gatecse-2022 algorithms hashing uniform-hashing one-mark

Answer key

1.16.6 Hashing: GATE CSE 2023 | Question: 10

An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and $k > m$.

Which one of the following is the best hashing strategy to counteract the adversary?

A. Division method, i.e., use the hash function $h(k) = k \bmod m$.
B. Multiplication method, i.e., use the hash function $h(k) = \lfloor m(kA - \lfloor kA \rfloor) \rfloor$, where A is a carefully chosen constant.
C. Universal hashing method.
D. If k is a prime number, use Division method. Otherwise, use Multiplication method.

gatecse-2023 algorithms hashing one-mark

Answer key

1.16.7 Hashing: GATE IT 2005 | Question: 16

A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key}\%10$. If the values $43, 165, 62, 123, 142$ are inserted in the table, in what location would the key value 142 be inserted?

A. 2 B. 3 C. 4 D. 6

gateit-2005 algorithms hashing easy

Answer key

1.17 Huffman Code (6)

1.17.1 Huffman Code: GATE CSE 1989 | Question: 13a

A language uses an alphabet of six letters, $\{a, b, c, d, e, f\}$. The relative frequency of use of each letter of the alphabet in the language is as given below:

LETTER	RELATIVE FREQUENCY OF USE
a	0.19
b	0.05
c	0.17
d	0.08
e	0.40
f	0.11

Design a prefix binary code for the language which would minimize the average length of the encoded words of the language.

descriptive gate1989 algorithms huffman-code

Answer key

1.17.2 Huffman Code: GATE CSE 2007 | Question: 76

Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

Which of the following is the Huffman code for the letter a, b, c, d, e, f?

A. $0, 10, 110, 1110, 11110, 11111$
B. $11, 10, 011, 010, 001, 000$
C. $11, 10, 01, 001, 0001, 0000$
D. $110, 100, 010, 000, 001, 111$

gatecse-2007 algorithms greedy-algorithms normal huffman-code

Answer key

1.17.3 Huffman Code: GATE CSE 2007 | Question: 77

Suppose the letters $a,\ b,\ c,\ d,\ e,\ f$ have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

What is the average length of the Huffman code for the letters $a,\ b,\ c,\ d,\ e,\ f$?

A. 3 B. 2.1875 C. 2.25 D. 1.9375

gatecse-2007 algorithms greedy-algorithms normal huffman-code

Answer key

1.17.4 Huffman Code: GATE CSE 2017 Set 2 | Question: 50

A message is made up entirely of characters from the set $X=\{P, Q, R, S, T\}$. The table of probabilities for each of the characters is shown below:

Character	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is ______.

gatecse-2017-set2 huffman-code numerical-answers algorithms

Answer key

1.17.5 Huffman Code: GATE CSE 2021 Set 2 | Question: 26

Consider the string abbccddeee. Each letter in the string must be assigned a binary code satisfying the following properties:

1. For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter.
2. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter.

Among the set of all binary code assignments which satisfy the above two properties, what is the minimum length of the encoded string?

A. 21 B. 23 C. 25 D. 30

gatecse-2021-set2 algorithms huffman-code two-marks

Answer key

1.17.6 Huffman Code: GATE IT 2006 | Question: 48

The characters a to h have the set of frequencies based on the first 8 Fibonacci numbers as follows

$a:1, b:1, c:2, d:3, e:5, f:8, g:13, h:21$

A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code?

110111100111010

A. $fdheg$ B. $ecgdf$ C. $dchfg$ D. $fehdg$

gateit-2006 algorithms greedy-algorithms normal huffman-code

Answer key

1.18 Identify Function (38)

1.18.1 Identify Function: GATE CSE 1989 | Question: 8a

What is the output produced by the following program, when the input is "HTGATE"

```
Function what (s:string): string;
var n:integer;
begin
    n = s.length
    if n <= 1
    then what := s
    else what :=contact (what (substring (s, 2, n)), s.C [1])
end;
```

Note

i. type string=record
   length:integer;
   C:array[1..100] of char
   end
ii. Substring (s, i, j): this yields the string made up of the i^{th} through j^{th} characters in s; for appropriately defined in i and j.
iii. Contact (s_1, s_2): this function yields a string of length s_1 length + s_2 - length obtained by concatenating s_1 with s_2 such that s_1 precedes s_2.

gate1989 descriptive algorithms identify-function

Answer key

1.18.2 Identify Function: GATE CSE 1990 | Question: 11b

The following program computes values of a mathematical function $f(x)$. Determine the form of $f(x)$.

```
main ()
{
   int m, n; float x, y, t;
   scanf ("%f%d", &x, &n);
   t = 1; y = 0; m = 1;
   do
   {
      t *= (-x/m);
      y += t;
   } while (m++ < n);
   printf ("The value of y is %f", y);
}
```

gate1990 descriptive algorithms identify-function

Answer key

1.18.3 Identify Function: GATE CSE 1991 | Question: 03-viii

Consider the following Pascal function:

```
Function X(M:integer):integer;
Var i:integer;
Begin
   i := 0;
   while i*i < M
   do i:= i+1
   X := i
 end
```

The function call $X(N)$, if N is positive, will return

A. $\lfloor\sqrt{N}\rfloor$
B. $\lfloor\sqrt{N}\rfloor + 1$
C. $\lceil\sqrt{N}\rceil$
D. $\lceil\sqrt{N}\rceil + 1$
E. None of the above

gate1991 algorithms easy identify-function multiple-selects

Answer key

1.18.4 Identify Function: GATE CSE 1993 | Question: 7.4

What does the following code do?

```
var a, b: integer;
begin
   a:=a+b;
   b:=a-b;
   a:=a-b;
end;
```

A. exchanges a and b
B. doubles a and stores in b
C. doubles b and stores in a
D. leaves a and b unchanged
E. none of the above

gate1993 algorithms identify-function easy

Answer key

1.18.5 Identify Function: GATE CSE 1994 | Question: 6

What function of x, n is computed by this program?

```
Function what(x, n:integer): integer:
Var
   value : integer
begin
   value := 1
   if n > 0 then
   begin
      if n mod 2 =1 then
         value := value * x;
      value := value * what(x*x, n div 2);
   end;
   what := value;
end;
```

gate1994 algorithms identify-function normal descriptive

Answer key

1.18.6 Identify Function: GATE CSE 1995 | Question: 1.4

In the following Pascal program segment, what is the value of X after the execution of the program segment?

```
X := -10; Y := 20;
If X > Y then if X < 0 then X := abs(X) else X := 2*X;
```

A. 10
B. -20
C. -10
D. None

gate1995 algorithms identify-function easy

Answer key

1.18.7 Identify Function: GATE CSE 1995 | Question: 2.3

Assume that X and Y are non-zero positive integers. What does the following Pascal program segment do?

```
while X <> Y do
if  X > Y then
   X := X - Y
else
   Y := Y - X;
write(X);
```

A. Computes the LCM of two numbers
B. Divides the larger number by the smaller number
C. Computes the GCD of two numbers
D. None of the above

gate1995 algorithms identify-function normal

Answer key

1.18.8 Identify Function: GATE CSE 1995 | Question: 4

A. Consider the following Pascal function where A and B are non-zero positive integers. What is the value of $\mathrm{GET}(3,2)$?

```
function GET(A,B:integer): integer;
begin
   if B=0 then
      GET:= 1
   else if A < B then
      GET:= 0
   else
      GET:= GET(A-1, B) + GET(A-1, B-1)
end;
```

B. The Pascal procedure given for computing the transpose of an $N \times N, (N > 1)$ matrix A of integers has an error. Find the error and correct it. Assume that the following declaration are made in the main program

```
const
   MAXSIZE=20;
type
   INTARR=array [1..MAXSIZE,1..MAXSIZE] of integer;
Procedure TRANSPOSE (var A: INTARR; N : integer);
var
   I, J, TMP: integer;
begin
   for I:=1 to N – 1 do
   for J:=1 to N do
   begin
      TMP:= A[I, J];
      A[I, J]:= A[J, I];
      A[J, I]:= TMP
   end
end;
```

gate1995 algorithms identify-function normal descriptive

Answer key

1.18.9 Identify Function: GATE CSE 1998 | Question: 2.12

What value would the following function return for the input $x = 95$?

```
Function fun (x:integer):integer;
Begin
  If x > 100 then fun = x – 10
  Else fun = fun(fun (x+11))
End;
```

A. 89
B. 90
C. 91
D. 92

gate1998 algorithms recursion identify-function normal

Answer key

1.18.10 Identify Function: GATE CSE 1999 | Question: 2.24

Consider the following C function definition

```
int Trial (int a, int b, int c)
{
  if ((a>=b) && (c<b)) return b;
  else if (a>=b) return Trial(a, c, b);
  else return Trial(b, a, c);
}
```

The functional Trial:

A. Finds the maximum of a, b, and c
B. Finds the minimum of a, b, and c
C. Finds the middle number of a, b, c
D. None of the above

gate1999 algorithms identify-function normal

Answer key

1.18.11 Identify Function: GATE CSE 2000 | Question: 2.15

Suppose you are given an array $s[1 \ldots n]$ and a procedure reverse (s, i, j) which reverses the order of elements in s between positions i and j (both inclusive). What does the following sequence do, where $1 \leqslant k \leqslant n$:

```
reverse (s, 1, k);
reverse (s, k+1, n);
reverse (s, 1, n);
```

A. Rotates s left by k positions
B. Leaves s unchanged
C. Reverses all elements of s
D. None of the above

gatecse-2000 algorithms normal identify-function

Answer key

1.18.12 Identify Function: GATE CSE 2003 | Question: 1

Consider the following C function.

For large values of y, the return value of the function f best approximates

```
float f,(float x, int y) {
    float p, s; int i;
    for (s=1,p=1,i=1; i<y; i++) {
        p *= x/i;
        s += p;
    }
    return s;
}
```

A. x^y
B. e^x
C. $\ln(1+x)$
D. x^x

gatecse-2003 algorithms identify-function normal

Answer key

1.18.13 Identify Function: GATE CSE 2003 | Question: 88

In the following C program fragment, j, k, n and TwoLog_n are integer variables, and A is an array of integers. The variable n is initialized to an integer $\geqslant 3$, and TwoLog_n is initialized to the value of $2^* \lceil \log_2(n) \rceil$

```
for (k = 3;  k <= n; k++)
        A[k] = 0;
for (k = 2; k <= TwoLog_n; k++)
    for (j = k+1; j <= n; j++)
        A[j] = A[j] || (j%k);
for (j = 3; j <= n; j++)
    if (!A[j]) printf("%d", j);
```

The set of numbers printed by this program fragment is

A. $\{m \mid m \leq n, (\exists i)\,[m = i!]\}$
B. $\{m \mid m \leq n, (\exists i)\,[m = i^2]\}$
C. $\{m \mid m \leq n, \text{m is prime}\}$
D. { }

gatecse-2003 algorithms identify-function normal

Answer key

1.18.14 Identify Function: GATE CSE 2004 | Question: 41

Consider the following C program

```
main()
{
    int x, y, m, n;
    scanf("%d %d", &x, &y);
    /* Assume x>0 and y>0*/
    m = x; n = y;
    while(m != n)
        {
            if (m > n)
```

```
            m = m-n;
        else
            n = n-m;
    }
    printf("%d", n);
}
```

The program computes

A. $x + y$ using repeated subtraction

B. $x \mod y$ using repeated subtraction

C. the greatest common divisor of x and y

D. the least common multiple of x and y

gatecse-2004 algorithms normal identify-function

Answer key

1.18.15 Identify Function: GATE CSE 2004 | Question: 42

What does the following algorithm approximate? (Assume $m > 1, \epsilon > 0$).

```
x = m;
y = 1;
While (x-y > ϵ)
{
    x = (x+y)/2;
    y = m/x;
}
print(x);
```

A. $\log m$

B. m^2

C. $m^{\frac{1}{2}}$

D. $m^{\frac{1}{3}}$

gatecse-2004 algorithms identify-function normal

Answer key

1.18.16 Identify Function: GATE CSE 2005 | Question: 31

Consider the following C-program:

```
void foo (int n, int sum) {
    int k = 0, j = 0;
    if (n == 0) return;
    k = n % 10; j = n/10;
    sum = sum + k;
    foo (j, sum);
    printf ("%d,",k);
}

int main() {
    int a = 2048, sum = 0;
    foo(a, sum);
    printf("%d\n", sum);
}
```

What does the above program print?

A. $8, 4, 0, 2, 14$

B. $8, 4, 0, 2, 0$

C. $2, 0, 4, 8, 14$

D. $2, 0, 4, 8, 0$

gatecse-2005 algorithms identify-function recursion normal

Answer key

1.18.17 Identify Function: GATE CSE 2006 | Question: 50

A set X can be represented by an array $x[n]$ as follows:

$$x\left[i\right] = \begin{cases} 1 & \text{if } i \in X \\ 0 & \text{otherwise} \end{cases}$$

Consider the following algorithm in which x, y, and z are Boolean arrays of size n:

```
algorithm zzz(x[], y[], z[]) {
int i;

for(i=0; i<n; ++i)
z[i] = (x[i]  ∧  ~y[i])  ∨  (~x[i]  ∧  y[i]);
```

```
}
```

The set Z computed by the algorithm is:

A. $(X \cup Y)$ B. $(X \cap Y)$ C. $(X-Y) \cap (Y-X)$ D. $(X-Y) \cup (Y-X)$

gatecse-2006 algorithms identify-function normal

Answer key

1.18.18 Identify Function: GATE CSE 2006 | Question: 53

Consider the following C-function in which $a[n]$ and $b[m]$ are two sorted integer arrays and $c[n+m]$ be another integer array,

```
void xyz(int a[], int b [], int c []){
   int i,j,k;
   i=j=k=0;
   while ((i<n) && (j<m))
      if (a[i] < b[j]) c[k++] = a[i++];
      else c[k++] = b[j++];
}
```

Which of the following condition(s) hold(s) after the termination of the while loop?

i. $j < m, k = n + j - 1$ and $a[n-1] < b[j]$ if $i = n$
ii. $i < n, k = m + i - 1$ and $b[m-1] \leq a[i]$ if $j = m$

A. only (i)
B. only (ii)
C. either (i) or (ii) but not both
D. neither (i) nor (ii)

gatecse-2006 algorithms identify-function normal

Answer key

1.18.19 Identify Function: GATE CSE 2009 | Question: 18

Consider the program below:

```
#include <stdio.h>
int fun(int n, int *f_p) {
   int t, f;
   if (n <= 1) {
      *f_p = 1;
      return 1;
   }
   t = fun(n-1, f_p);
   f = t + *f_p;
   *f_p = t;
   return f;
}

int main() {
   int x = 15;
   printf("%d/n", fun(5, &x));
   return 0;
}
```

The value printed is:

A. 6 B. 8 C. 14 D. 15

gatecse-2009 algorithms recursion identify-function normal

Answer key

1.18.20 Identify Function: GATE CSE 2010 | Question: 35

What is the value printed by the following C program?

```
#include<stdio.h>

int f(int *a, int n)
{
   if (n <= 0) return 0;
   else if (*a % 2 == 0) return *a+f(a+1, n-1);
   else return *a - f(a+1, n-1);
```

```
}

int main()
{
    int a[] = {12, 7, 13, 4, 11, 6};
    printf("%d", f(a, 6));
    return 0;
}
```

A. -9 B. 5 C. 15 D. 19

gatecse-2010 algorithms recursion identify-function normal

Answer key

1.18.21 Identify Function: GATE CSE 2011 | Question: 48

Consider the following recursive C function that takes two arguments.

```
unsigned int foo(unsigned int n, unsigned int r) {
    if (n>0) return ((n%r) + foo(n/r, r));
    else return 0;
}
```

What is the return value of the function foo when it is called as $\text{foo}(345, 10)$?

A. 345 B. 12 C. 5 D. 3

gatecse-2011 algorithms recursion identify-function normal

Answer key

1.18.22 Identify Function: GATE CSE 2011 | Question: 49

Consider the following recursive C function that takes two arguments.

```
unsigned int foo(unsigned int n, unsigned int r) {
    if (n>0) return ((n%r) + foo(n/r, r));
    else return 0;
}
```

What is the return value of the function foo when it is called as $\text{foo}(513, 2)$?

A. 9 B. 8 C. 5 D. 2

gatecse-2011 algorithms recursion identify-function normal

Answer key

1.18.23 Identify Function: GATE CSE 2013 | Question: 31

Consider the following function:

```
int unknown(int n){

int i, j, k=0;
for (i=n/2; i<=n; i++)
    for (j=2; j<=n; j=j*2)
        k = k + n/2;
 return (k);

}
```

The return value of the function is

A. $\Theta(n^2)$ B. $\Theta(n^2 \log n)$
C. $\Theta(n^3)$ D. $\Theta(n^3 \log n)$

gatecse-2013 algorithms identify-function normal

Answer key

1.18.24 Identify Function: GATE CSE 2014 Set 1 | Question: 41

Consider the following C function in which **size** is the number of elements in the array **E**:

```
int MyX(int *E, unsigned int size)
```

```
{
  int Y = 0;
  int Z;
  int i, j, k;

  for(i = 0; i< size; i++)
        Y = Y + E[i];

   for(i=0; i < size; i++)
     for(j = i; j < size; j++)
     {
         Z = 0;
         for(k = i; k <= j; k++)
            Z = Z + E[k];
         if(Z > Y)
            Y = Z;
     }
  return Y;
}
```

The value returned by the function **MyX** is the

A. maximum possible sum of elements in any sub-array of array **E**.
B. maximum element in any sub-array of array **E.**
C. sum of the maximum elements in all possible sub-arrays of array **E**.
D. the sum of all the elements in the array **E.**

gatecse-2014-set1 algorithms identify-function normal

Answer key

1.18.25 Identify Function: GATE CSE 2014 Set 2 | Question: 10

Consider the function func shown below:

```
int func(int num) {
  int count = 0;
  while (num) {
    count++;
    num>>= 1;
  }
  return (count);
}
```

The value returned by func(435) is ________

gatecse-2014-set2 algorithms identify-function numerical-answers easy

Answer key

1.18.26 Identify Function: GATE CSE 2014 Set 3 | Question: 10

Let A be the square matrix of size $n \times n$. Consider the following pseudocode. What is the expected output?

```
C=100;
for i=1 to n do
   for j=1 to n do
   {
      Temp = A[i][j]+C;
      A[i][j] = A[j][i];
      A[j][i] = Temp -C;
   }
for i=1 to n do
   for j=1 to n do
      output (A[i][j]);
```

A. The matrix A itself
B. Transpose of the matrix A
C. Adding 100 to the upper diagonal elements and subtracting 100 from lower diagonal elements of A
D. None of the above

gatecse-2014-set3 algorithms identify-function easy

Answer key

1.18.27 Identify Function: GATE CSE 2015 Set 1 | Question: 31

Consider the following C function.

```
int fun1 (int n) {
    int i, j, k, p, q = 0;
    for (i = 1; i < n; ++i)
    {
      p = 0;
     for (j = n; j > 1; j = j/2)
        ++p;
     for (k = 1; k < p; k = k * 2)
        ++q;
    }
    return q;
}
```

Which one of the following most closely approximates the return value of the function fun1?

A. n^3
B. $n(\log n)^2$
C. $n \log n$
D. $n \log(\log n)$

gatecse-2015-set1 algorithms normal identify-function

Answer key

1.18.28 Identify Function: GATE CSE 2015 Set 2 | Question: 11

Consider the following C function.

```
int fun(int n) {
   int x=1, k;
   if (n==1) return x;
   for (k=1; k<n; ++k)
      x = x + fun(k) * fun (n-k);
   return x;
}
```

The return value of $fun(5)$ is ______.

gatecse-2015-set2 algorithms identify-function recurrence-relation normal numerical-answers

Answer key

1.18.29 Identify Function: GATE CSE 2015 Set 3 | Question: 49

Suppose $c = \langle c[0], \dots, c[k-1] \rangle$ is an array of length k, where all the entries are from the set $\{0, 1\}$. For any positive integers a and n, consider the following pseudocode.

```
DOSOMETHING (c, a, n)

z ← 1
for i ← 0 to k-1
    do z ← z² mod n
    if c[i]=1
        then z ← (z × a) mod n
return z
```

If $k = 4, c = \langle 1, 0, 1, 1 \rangle, a = 2,$ and $n = 8$, then the output of DOSOMETHING(*c, a, n)* is _______.

gatecse-2015-set3 algorithms identify-function normal numerical-answers

Answer key

1.18.30 Identify Function: GATE CSE 2019 | Question: 26

Consider the following C function.

```
void convert (int n ) {
      if (n<0)
         printf(“%d”, n);
      else {
         convert(n/2);
         printf(“%d”, n%2);
```

```
        }
}
```

Which one of the following will happen when the function *convert* is called with any positive integer n as argument?

A. It will print the binary representation of n and terminate
B. It will print the binary representation of n in the reverse order and terminate
C. It will print the binary representation of n but will not terminate
D. It will not print anything and will not terminate

gatecse-2019 algorithms identify-function two-marks

Answer key

1.18.31 Identify Function: GATE CSE 2020 | Question: 48

Consider the following C functions.

```c
int tob (int b, int* arr) {
    int i;
    for (i = 0; b>0; i++) {
        if (b%2)  arr [i] = 1;
        else      arr[i] = 0;
        b = b/2;
    }
    return (i);
}
```

```c
int pp(int a, int b) {
    int  arr[20];
    int i, tot = 1, ex, len;
    ex = a;
    len = tob(b, arr);
    for (i=0; i<len ; i++) {
        if (arr[i] ==1)
            tot = tot * ex;
        ex= ex*ex;
    }
    return (tot) ;
}
```

The value returned by $pp(3, 4)$ is _______.

gatecse-2020 numerical-answers identify-function two-marks

Answer key

1.18.32 Identify Function: GATE CSE 2021 Set 1 | Question: 48

Consider the following $\mathrm{ANSI\ C}$ function:

```c
int SimpleFunction(int Y[], int n, int x)
{
    int total = Y[0], loopIndex;
    for (loopIndex=1; loopIndex<=n-1; loopIndex++)
        total=x*total +Y[loopIndex];
    return total;
}
```

Let Z be an array of 10 elements with $\mathsf{Z}[i] = 1$, for all i such that $0 \leq i \leq 9$. The value returned by $\mathsf{SimpleFunction}(\mathsf{Z}, 10, 2)$ is __________

gatecse-2021-set1 algorithms numerical-answers identify-function two-marks

Answer key

1.18.33 Identify Function: GATE CSE 2021 Set 2 | Question: 23

Consider the following $\mathrm{ANSI\ C}$ function:

```c
int SomeFunction (int x, int y)
{
    if ((x==1) || (y==1)) return 1;
    if (x==y) return x;
    if (x > y) return SomeFunction(x-y, y);
    if (y > x) return SomeFunction (x, y-x);

}
```

The value returned by $\mathrm{SomeFunction}(15, 255)$ is __________ gatecse-2021-set2 numerical-answers algorithms identify-function output one-mark

Answer key

1.18.34 Identify Function: GATE IT 2005 | Question: 53

The following C function takes two ASCII strings and determines whether one is an anagram of the other. An anagram of a string s is a string obtained by permuting the letters in s.

```
int anagram (char *a, char *b) {
   int count [128], j;
   for (j = 0;  j < 128; j++) count[j] = 0;
   j = 0;
   while (a[j] && b[j]) {
      A;
      B;
   }
   for (j = 0; j < 128; j++) if (count [j]) return 0;
   return 1;
}
```

Choose the correct alternative for statements A and B.

A. A: count [a[j]]++ and B: count[b[j]]--
B. A: count [a[j]]++ and B: count[b[j]]++
C. A: count [a[j++]]++ and B: count[b[j]]--
D. A: count [a[j]]++ and B: count[b[j++]]--

gateit-2005 normal identify-function

Answer key

1.18.35 Identify Function: GATE IT 2005 | Question: 57

What is the output printed by the following program?

```
#include <stdio.h>

int f(int n, int k) {
   if (n == 0) return 0;
   else if (n % 2) return f(n/2, 2*k) + k;
   else return f(n/2, 2*k) - k;
}

int main () {
   printf("%d", f(20, 1));
   return 0;
}
```

A. 5
B. 8
C. 9
D. 20

gateit-2005 algorithms identify-function normal

Answer key

1.18.36 Identify Function: GATE IT 2006 | Question: 52

The following function computes the value of $\binom{m}{n}$ correctly for all legal values m and n ($m \geq 1, n \geq 0$ and $m > n$)

```
int func(int m, int n)
{
   if (E) return 1;
   else return(func(m -1, n) + func(m - 1, n - 1));
}
```

In the above function, which of the following is the correct expression for E?

A. $(n == 0) || (m == 1)$
B. $(n == 0)$ && $(m == 1)$
C. $(n == 0) || (m == n)$
D. $(n == 0)$ && $(m == n)$

gateit-2006 algorithms identify-function normal

Answer key

1.18.37 Identify Function: GATE IT 2008 | Question: 82

Consider the code fragment written in C below :

```
void f (int n)
{
 if (n <=1) {
  printf ("%d", n);
 }
 else {
  f (n/2);
  printf ("%d", n%2);
 }
}
```

What does f(173) print?

A. 010110101 B. 010101101 C. 10110101 D. 10101101

gateit-2008 algorithms recursion identify-function normal

Answer key

1.18.38 Identify Function: GATE IT 2008 | Question: 83

Consider the code fragment written in C below :

```
void f (int n)
{
    if (n <= 1) {
        printf ("%d", n);
    }
    else {
        f (n/2);
        printf ("%d", n%2);
    }
}
```

Which of the following implementations will produce the same output for $f(173)$ as the above code?

P1

```
void f (int n)
{
    if (n/2) {
        f(n/2);
    }
    printf ("%d", n%2);
}
```

P2

```
void f (int n)
{
    if (n <=1) {
        printf ("%d", n);
    }
    else {
        printf ("%d", n%2);
        f (n/2);
    }
}
```

A. Both $P1$ and $P2$ B. $P2$ only C. $P1$ only D. Neither $P1$ nor $P2$

gateit-2008 algorithms recursion identify-function normal

Answer key

1.19 Insertion Sort (3)

1.19.1 Insertion Sort: GATE CSE 2003 | Question: 22

The usual $\Theta(n^2)$ implementation of Insertion Sort to sort an array uses linear search to identify the position where an element is to be inserted into the already sorted part of the array. If, instead, we use binary search to identify the position, the worst case running time will

A. remain $\Theta(n^2)$
B. become $\Theta(n(\log n)^2)$
C. become $\Theta(n \log n)$
D. become $\Theta(n)$

gatecse-2003 algorithms sorting time-complexity normal insertion-sort

Answer key

1.19.2 Insertion Sort: GATE CSE 2003 | Question: 62

In a permutation $a_1 \dots a_n$, of n distinct integers, an inversion is a pair (a_i, a_j) such that $i < j$ and $a_i > a_j$. What would be the worst case time complexity of the Insertion Sort algorithm, if the inputs are restricted to permutations of $1 \dots n$ with at most n inversions?

A. $\Theta(n^2)$ B. $\Theta(n \log n)$ C. $\Theta(n^{1.5})$ D. $\Theta(n)$

gatecse-2003 algorithms sorting normal insertion-sort

Answer key

1.19.3 Insertion Sort: GATE DA 2025 | Question: 19

Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x.

A. 10 B. 12 C. 14 D. 16

gateda-2025 algorithms insertion-sort sorting multiple-selects easy one-mark

Answer key

1.20 Linear Probing (1)

1.20.1 Linear Probing: GATE DA 2025 | Question: 8

Consider a hash table of size 10 with indices $\{0, 1, \dots, 9\}$, with the hash function

$$h(x) = 3x \pmod{10}$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: $1, 4, 5, 6, 14, 15$. The indices where the keys 14 and 15 are stored are, respectively

A. 2 and 5 B. 2 and 6 C. 4 and 5 D. 4 and 6

gateda-2025 algorithms hashing linear-probing easy one-mark

Answer key

1.21 Matrix Chain Ordering (3)

1.21.1 Matrix Chain Ordering: GATE CSE 2011 | Question: 38

Four Matrices M_1, M_2, M_3 and M_4 of dimensions $p \times q,\ q \times r,\ r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of scalar multiplications is $pqr + rst + prt$. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is $pqr + prs + pst$.

If $p = 10, q = 100, r = 20, s = 5$ and $t = 80$, then the minimum number of scalar multiplications needed is

A. 248000 B. 44000 C. 19000 D. 25000

gatecse-2011 algorithms dynamic-programming normal matrix-chain-ordering

Answer key

1.21.2 Matrix Chain Ordering: GATE CSE 2016 Set 2 | Question: 38

Let A_1, A_2, A_3 and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$ and 10×5, respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _________.

gatecse-2016-set2 dynamic-programming algorithms matrix-chain-ordering normal numerical-answers

Answer key

1.21.3 Matrix Chain Ordering: GATE CSE 2018 | Question: 31

Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1 G_2 G_3 \dots G_n$ can be done by parenthesizing in different ways. Define $G_i G_{i+1}$ as an **explicitly computed pair** for a given paranthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1 G_2 G_3 G_4 G_5 G_6$ using parenthesization $(G_1(G_2G_3))(G_4(G_5G_6))$, G_2G_3 and G_5G_6 are only explicitly computed pairs.

Consider a matrix multiplication chain $F_1F_2F_3F_4F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are of dimensions $2 \times 25, 25 \times 3, 3 \times 16, 16 \times 1$ and 1×1000, respectively. In the parenthesization of $F_1F_2F_3F_4F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

A. F_1F_2 and F_3F_4 only
B. F_2F_3 only
C. F_3F_4 only
D. F_1F_2 and F_4F_5 only

gatecse-2018 algorithms dynamic-programming two-marks matrix-chain-ordering

Answer key

1.22 Merge Sort (3)

1.22.1 Merge Sort: GATE CSE 1999 | Question: 1.14, ISRO2015-42

If one uses straight two-way merge sort algorithm to sort the following elements in ascending order:

$20,\ 47,\ 15,\ 8,\ 9,\ 4,\ 40,\ 30,\ 12,\ 17$

then the order of these elements after second pass of the algorithm is:

A. $8,\ 9,\ 15,\ 20,\ 47,\ 4,\ 12,\ 17,\ 30,\ 40$
B. $8,\ 15,\ 20,\ 47,\ 4,\ 9,\ 30,\ 40,\ 12,\ 17$
C. $15,\ 20,\ 47,\ 4,\ 8,\ 9,\ 12,\ 30,\ 40,\ 17$
D. $4,\ 8,\ 9,\ 15,\ 20,\ 47,\ 12,\ 17,\ 30,\ 40$

gate1999 algorithms merge-sort normal isro2015 sorting

Answer key

1.22.2 Merge Sort: GATE CSE 2012 | Question: 39

A list of n strings, each of length n, is sorted into lexicographic order using the merge-sort algorithm. The worst case running time of this computation is

A. $O(n \log n)$
B. $O(n^2 \log n)$
C. $O(n^2 + \log n)$
D. $O(n^2)$

gatecse-2012 algorithms sorting normal merge-sort

Answer key

1.22.3 Merge Sort: GATE CSE 2015 Set 3 | Question: 27

Assume that a mergesort algorithm in the worst case takes 30 seconds for an input of size 64. Which of the following most closely approximates the maximum input size of a problem that can be solved in 6 minutes?

A. 256
B. 512
C. 1024
D. 2018

gatecse-2015-set3 algorithms sorting merge-sort

Answer key

1.23 Merging (2)

1.23.1 Merging: GATE CSE 1995 | Question: 1.16

For merging two sorted lists of sizes m and n into a sorted list of size $m + n$, we require comparisons of

A. $O(m)$
B. $O(n)$
C. $O(m + n)$
D. $O(\log m + \log n)$

gate1995 algorithms sorting normal merging

Answer key

1.23.2 Merging: GATE CSE 2014 Set 2 | Question: 38

Suppose P, Q, R, S, T are sorted sequences having lengths $20, 24, 30, 35, 50$ respectively. They are to be merged into a single sequence by merging together two sequences at a time. The number of comparisons that will be needed in the worst case by the optimal algorithm for doing this is ____.

gatecse-2014-set2 algorithms sorting normal numerical-answers merging

Answer key

1.24 Minimum Spanning Tree (34)

1.24.1 Minimum Spanning Tree: GATE CSE 1991 | Question: 03,vi

Kruskal's algorithm for finding a minimum spanning tree of a weighted graph G with n vertices and m edges has the time complexity of:

A. $O(n^2)$
B. $O(mn)$
C. $O(m+n)$
D. $O(m \log n)$
E. $O(m^2)$

gate1991 algorithms graph-algorithms minimum-spanning-tree time-complexity multiple-selects

Answer key

1.24.2 Minimum Spanning Tree: GATE CSE 1992 | Question: 01,ix

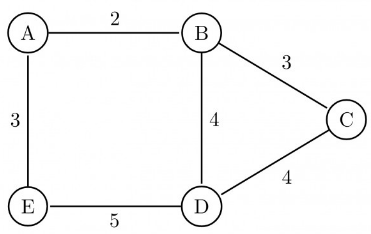
Complexity of Kruskal's algorithm for finding the minimum spanning tree of an undirected graph containing n vertices and m edges if the edges are sorted is _______

gate1992 minimum-spanning-tree algorithms time-complexity easy fill-in-the-blanks

Answer key

1.24.3 Minimum Spanning Tree: GATE CSE 1995 | Question: 22

How many minimum spanning trees does the following graph have? Draw them. (Weights are assigned to edges).



gate1995 algorithms graph-algorithms minimum-spanning-tree easy descriptive

Answer key

1.24.4 Minimum Spanning Tree: GATE CSE 1996 | Question: 16

A complete, undirected, weighted graph G is given on the vertex $\{0, 1, \ldots, n-1\}$ for any fixed 'n'. Draw the minimum spanning tree of G if

A. the weight of the edge (u, v) is $|u - v|$
B. the weight of the edge (u, v) is $u + v$

gate1996 algorithms graph-algorithms minimum-spanning-tree normal descriptive

Answer key

1.24.5 Minimum Spanning Tree: GATE CSE 1997 | Question: 9

Consider a graph whose vertices are points in the plane with integer co-ordinates (x, y) such that $1 \leq x \leq n$ and $1 \leq y \leq n$, where $n \geq 2$ is an integer. Two vertices (x_1, y_1) and (x_2, y_2) are adjacent iff $|x_1 - x_2| \leq 1 \text{ and } | y_1 - y_2 | \leq 1$. The weight of an edge $\{(x_1, y_1), (x_2, y_2)\}$ is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

A. What is the weight of a minimum weight-spanning tree in this graph? Write only the answer without any explanations.

B. What is the weight of a maximum weight-spanning tree in this graph? Write only the answer without any explanations.

gate1997 algorithms minimum-spanning-tree normal descriptive

Answer key

1.24.6 Minimum Spanning Tree: GATE CSE 2000 | Question: 2.18

Let G be an undirected connected graph with distinct edge weights. Let e_{max} be the edge with maximum weight and e_{min} the edge with minimum weight. Which of the following statements is false?

A. Every minimum spanning tree of G must contain e_{min}
B. If e_{max} is in a minimum spanning tree, then its removal must disconnect G
C. No minimum spanning tree contains e_{max}
D. G has a unique minimum spanning tree

gatecse-2000 algorithms minimum-spanning-tree normal

Answer key

1.24.7 Minimum Spanning Tree: GATE CSE 2001 | Question: 15

Consider a weighted undirected graph with vertex set $V = \{n1, n2, n3, n4, n5, n6\}$ and edge set $E = \{(n1, n2, 2), (n1, n3, 8), (n1, n6, 3), (n2, n4, 4), (n2, n5, 12), (n3, n4, 7), (n4, n5, 9), (n4, n6, 4)\}$.

The third value in each tuple represents the weight of the edge specified in the tuple.

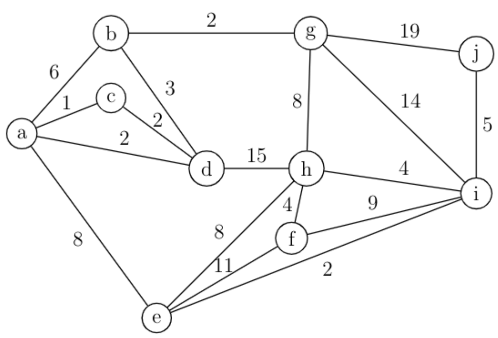
A. List the edges of a minimum spanning tree of the graph.
B. How many distinct minimum spanning trees does this graph have?
C. Is the minimum among the edge weights of a minimum spanning tree unique over all possible minimum spanning trees of a graph?
D. Is the maximum among the edge weights of a minimum spanning tree unique over all possible minimum spanning tree of a graph?

gatecse-2001 algorithms minimum-spanning-tree normal descriptive

Answer key

1.24.8 Minimum Spanning Tree: GATE CSE 2003 | Question: 68

What is the weight of a minimum spanning tree of the following graph?



A. 29 B. 31 C. 38 D. 41

gatecse-2003 algorithms minimum-spanning-tree normal

Answer key

1.24.9 Minimum Spanning Tree: GATE CSE 2005 | Question: 6

An undirected graph G has n nodes. its adjacency matrix is given by an $n \times n$ square matrix whose (i) diagonal elements are 0's and (ii) non-diagonal elements are 1's. Which one of the following is TRUE?

A. Graph G has no minimum spanning tree (MST)
B. Graph G has unique MST of cost $n-1$
C. Graph G has multiple distinct MSTs, each of cost $n-1$
D. Graph G has multiple spanning trees of different costs

gatecse-2005 algorithms minimum-spanning-tree normal

Answer key

1.24.10 Minimum Spanning Tree: GATE CSE 2006 | Question: 11

Consider a weighted complete graph G on the vertex set $\{v_1, v_2, \dots . . v_n\}$ such that the weight of the edge (v_i, v_j) is $2|i-j|$. The weight of a minimum spanning tree of G is:

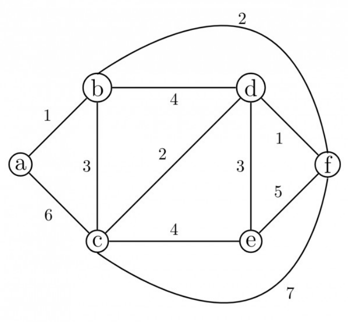
A. $n-1$ B. $2n-2$ C. $\binom{n}{2}$ D. n^2

gatecse-2006 algorithms minimum-spanning-tree normal

Answer key

1.24.11 Minimum Spanning Tree: GATE CSE 2006 | Question: 47

Consider the following graph:



Which one of the following cannot be the sequence of edges added, **in that order**, to a minimum spanning tree using Kruskal's algorithm?

A. $(a-b),(d-f),(b-f),(d-c),(d-e)$
B. $(a-b),(d-f),(d-c),(b-f),(d-e)$
C. $(d-f),(a-b),(d-c),(b-f),(d-e)$
D. $(d-f),(a-b),(b-f),(d-e),(d-c)$

gatecse-2006 algorithms graph-algorithms minimum-spanning-tree normal

Answer key

1.24.12 Minimum Spanning Tree: GATE CSE 2007 | Question: 49

Let w be the minimum weight among all edge weights in an undirected connected graph. Let e be a specific edge of weight w. Which of the following is FALSE?

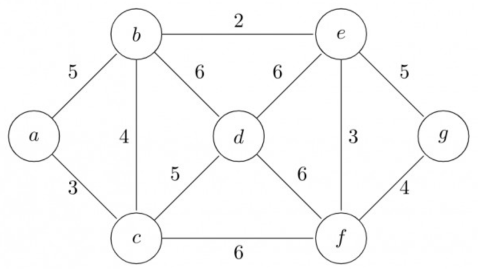
A. There is a minimum spanning tree containing e
B. If e is not in a minimum spanning tree T, then in the cycle formed by adding e to T, all edges have the same weight.
C. Every minimum spanning tree has an edge of weight w
D. e is present in every minimum spanning tree

gatecse-2007 algorithms minimum-spanning-tree normal

Answer key

1.24.13 Minimum Spanning Tree: GATE CSE 2009 | Question: 38

Consider the following graph:



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?

A. (b, e) (e, f) (a, c) (b, c) (f, g) (c, d)
B. (b, e) (e, f) (a, c) (f, g) (b, c) (c, d)
C. (b, e) (a, c) (e, f) (b, c) (f, g) (c, d)
D. (b, e) (e, f) (b, c) (a, c) (f, g) (c, d)

gatecse-2009 algorithms minimum-spanning-tree normal

Answer key

1.24.14 Minimum Spanning Tree: GATE CSE 2010 | Question: 50

Consider a complete undirected graph with vertex set $\{0, 1, 2, 3, 4\}$. Entry W_{ij} in the matrix W below is the weight of the edge $\{i, j\}$

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix}$$

What is the minimum possible weight of a spanning tree T in this graph such that vertex 0 is a leaf node in the tree T?

A. 7 B. 8 C. 9 D. 10

gatecse-2010 algorithms minimum-spanning-tree normal

Answer key

1.24.15 Minimum Spanning Tree: GATE CSE 2010 | Question: 51

Consider a complete undirected graph with vertex set $\{0, 1, 2, 3, 4\}$. Entry W_{ij} in the matrix W below is the weight of the edge $\{i, j\}$

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix}$$

What is the minimum possible weight of a path P from vertex 1 to vertex 2 in this graph such that P contains at most 3 edges?

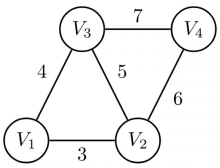
A. 7 B. 8 C. 9 D. 10

gatecse-2010 normal algorithms minimum-spanning-tree

Answer key

1.24.16 Minimum Spanning Tree: GATE CSE 2011 | Question: 54

An undirected graph $G(V, E)$ contains $n\ (n > 2)$ nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i, v_j are connected if and only if $0 <| i - j | \leq 2$. Each edge (v_i, v_j) is assigned a weight $i + j$. A sample graph with $n = 4$ is shown below.



What will be the cost of the minimum spanning tree (MST) of such a graph with n nodes?

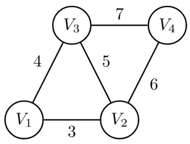
A. $\frac{1}{12}(11n^2 - 5n)$ B. $n^2 - n + 1$ C. $6n - 11$ D. $2n + 1$

gatecse-2011 algorithms graph-algorithms minimum-spanning-tree normal

Answer key

1.24.17 Minimum Spanning Tree: GATE CSE 2011 | Question: 55

An undirected graph $G(V, E)$ contains $n\ (n > 2)$ nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i, v_j are connected if and only if $0 <| i - j | \leq 2$. Each edge (v_i, v_j) is assigned a weight $i + j$. A sample graph with $n = 4$ is shown below.



The length of the path from v_5 to v_6 in the MST of previous question with $n = 10$ is

A. 11 B. 25 C. 31 D. 41

gatecse-2011 algorithms graph-algorithms minimum-spanning-tree normal

Answer key

1.24.18 Minimum Spanning Tree: GATE CSE 2012 | Question: 29

Let G be a weighted graph with edge weights greater than one and G' be the graph constructed by squaring the weights of edges in G. Let T and T' be the minimum spanning trees of G and G', respectively, with total weights t and t'. Which of the following statements is **TRUE**?

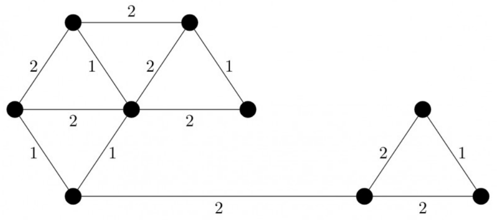
A. $T' = T$ with total weight $t' = t^2$
B. $T' = T$ with total weight $t' < t^2$
C. $T' \neq T$ but total weight $t' = t^2$
D. None of the above

gatecse-2012 algorithms minimum-spanning-tree normal marks-to-all

Answer key

1.24.19 Minimum Spanning Tree: GATE CSE 2014 Set 2 | Question: 52

The number of distinct minimum spanning trees for the weighted graph below is _____

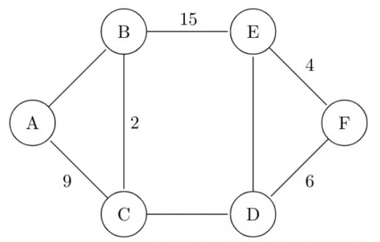


gatecse-2014-set2 algorithms minimum-spanning-tree numerical-answers normal

Answer key☟

1.24.20 Minimum Spanning Tree: GATE CSE 2015 Set 1 | Question: 43

The graph shown below has 8 edges with distinct integer edge weights. The minimum spanning tree (**MST**) is of weight 36 and contains the edges: $\{(A, C), (B, C), (B, E), (E, F), (D, F)\}$. The edge weights of only those edges which are in the **MST** are given in the figure shown below. The minimum possible sum of weights of all 8 edges of this graph is_______________.



gatecse-2015-set1 algorithms minimum-spanning-tree normal numerical-answers

Answer key☟

1.24.21 Minimum Spanning Tree: GATE CSE 2015 Set 3 | Question: 40

Let G be a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes ______.

gatecse-2015-set3 algorithms minimum-spanning-tree easy numerical-answers

Answer key☟

1.24.22 Minimum Spanning Tree: GATE CSE 2016 Set 1 | Question: 14

Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE?

- P: Minimum spanning tree of G does not change.
- Q: Shortest path between any pair of vertices does not change.

A. P only
B. Q only
C. Neither P nor Q
D. Both P and Q

gatecse-2016-set1 algorithms minimum-spanning-tree normal

Answer key☟

1.24.23 Minimum Spanning Tree: GATE CSE 2016 Set 1 | Question: 39

Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being $1, 2, 3, 4, 5,$ and 6. The maximum possible weight that a minimum weight spanning tree of G can have is __________

gatecse-2016-set1 algorithms minimum-spanning-tree normal numerical-answers

Answer key☟

1.24.24 Minimum Spanning Tree: GATE CSE 2016 Set 1 | Question: 40

$G = (V, E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G. Which of the following statements about the minimum spanning trees $(MSTs)$ of G is/are TRUE?

I. If e is the lightest edge of some cycle in G, then every MST of G includes e.
II. If e is the heaviest edge of some cycle in G, then every MST of G excludes e.

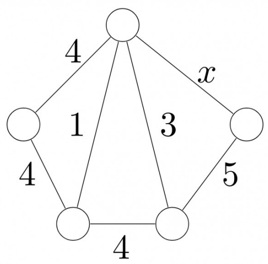
A. I only. B. II only. C. Both I and II. D. Neither I nor II.

gatecse-2016-set1 algorithms minimum-spanning-tree normal

Answer key

1.24.25 Minimum Spanning Tree: GATE CSE 2018 | Question: 47

Consider the following undirected graph G:



Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G. The number of MWSTs of G for this value of x is ____.

gatecse-2018 algorithms graph-algorithms minimum-spanning-tree numerical-answers two-marks

Answer key

1.24.26 Minimum Spanning Tree: GATE CSE 2020 | Question: 31

Let $G = (V, E)$ be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighed edge $(u, v) \in V \times V$ is added to G. The worst case time complexity of determining if T is still an MST of the resultant graph is

A. $\Theta(\mid E \mid + \mid V \mid)$

B. $\Theta(\mid E \mid\mid V \mid)$

C. $\Theta(E \mid \log \mid V \mid)$

D. $\Theta(\mid V \mid)$

gatecse-2020 algorithms minimum-spanning-tree graph-algorithms two-marks

Answer key

1.24.27 Minimum Spanning Tree: GATE CSE 2020 | Question: 49

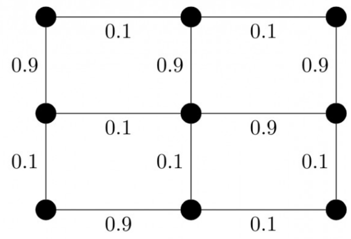
Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_{100}\}$, $E = \{(v_i, v_j) \mid 1 \leq i < j \leq 100\}$, and weight of the edge (v_i, v_j) is $|i - j|$. The weight of minimum spanning tree of G is _________

gatecse-2020 numerical-answers algorithms graph-algorithms two-marks minimum-spanning-tree

Answer key

1.24.28 Minimum Spanning Tree: GATE CSE 2021 Set 1 | Question: 17

Consider the following undirected graph with edge weights as shown:



The number of minimum-weight spanning trees of the graph is ___________.

gatecse-2021-set1 algorithms graph-algorithms minimum-spanning-tree numerical-answers one-mark

Answer key☟

1.24.29 Minimum Spanning Tree: GATE CSE 2021 Set 2 | Question: 1

Let G be a connected undirected weighted graph. Consider the following two statements.

- S_1: There exists a minimum weight edge in G which is present in every minimum spanning tree of G.
- S_2: If every edge in G has distinct weight, then G has a unique minimum spanning tree.

Which one of the following options is correct?

A. Both S_1 and S_2 are true
B. S_1 is true and S_2 is false
C. S_1 is false and S_2 is true
D. Both S_1 and S_2 are false

gatecse-2021-set2 algorithms graph-algorithms minimum-spanning-tree one-mark

Answer key☟

1.24.30 Minimum Spanning Tree: GATE CSE 2022 | Question: 39

Consider a simple undirected weighted graph G, all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?

A. The edge with the second smallest weight is always part of any minimum spanning tree of G.
B. One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G.
C. Suppose $S \subseteq V$ be such that $S \neq \phi$ and $S \neq V$. Consider the edge with the minimum weight such that one of its vertices is in S and the other in $V \setminus S$. Such an edge will always be part of any minimum spanning tree of G.
D. G can have multiple minimum spanning trees.

gatecse-2022 algorithms minimum-spanning-tree multiple-selects two-marks

Answer key☟

1.24.31 Minimum Spanning Tree: GATE CSE 2022 | Question: 48

Let $G(V,E)$ be a directed graph, where $V = \{1,2,3,4,5\}$ is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A.

$$A[i][j] = \begin{cases} 1, & 1 \leq j \leq i \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

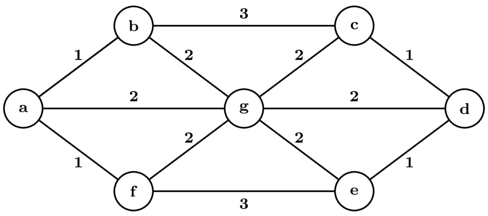
$A[i][j] = 1$ indicates a directed edge from node i to node j. A $directed\ spanning\ tree$ of G, rooted at $r \in V$, is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V. The number of such directed spanning trees rooted at vertex 5 is ___________________.

gatecse-2022 numerical-answers algorithms minimum-spanning-tree two-marks

Answer key☟

1.24.32 Minimum Spanning Tree: GATE CSE 2024 | Set 2 | Question: 49

The number of distinct minimum-weight spanning trees of the following graph is

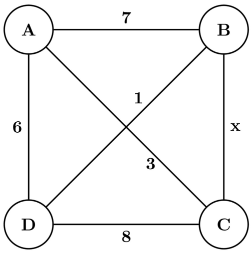


gatecse2024-set2 numerical-answers algorithms minimum-spanning-tree two-marks

Answer key☟

1.24.33 Minimum Spanning Tree: GATE CSE 2025 | Set 1 | Question: 54

The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is __________. (answer in integer)



gatecse2025-set1 algorithms minimum-spanning-tree numerical-answers easy two-marks

Answer key☟

1.24.34 Minimum Spanning Tree: GATE IT 2005 | Question: 52

Let G be a weighted undirected graph and e be an edge with maximum weight in G. Suppose there is a minimum weight spanning tree in G containing the edge e. Which of the following statements is always TRUE?

A. There exists a cutset in G having all edges of maximum weight.
B. There exists a cycle in G having all edges of maximum weight.
C. Edge e cannot be contained in a cycle.
D. All edges in G have the same weight.

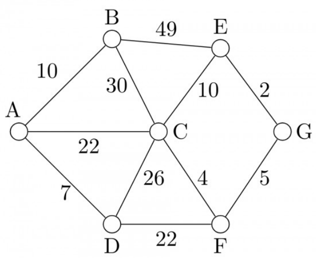
gateit-2005 algorithms minimum-spanning-tree normal

Answer key☟

1.25 Prims Algorithm (2)

1.25.1 Prims Algorithm: GATE IT 2004 | Question: 56

Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

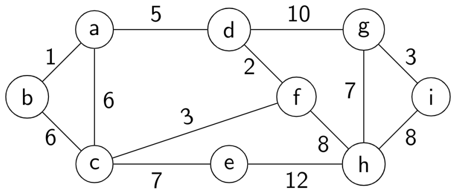
A. $(\mathrm{E}, \mathrm{G}), (\mathrm{C}, \mathrm{F}), (\mathrm{F}, \mathrm{G}), (\mathrm{A}, \mathrm{D}), (\mathrm{A}, \mathrm{B}), (\mathrm{A}, \mathrm{C})$
B. $(\mathrm{A}, \mathrm{D}), (\mathrm{A}, \mathrm{B}), (\mathrm{A}, \mathrm{C}), (\mathrm{C}, \mathrm{F}), (\mathrm{G}, \mathrm{E}), (\mathrm{F}, \mathrm{G})$
C. $(\mathrm{A}, \mathrm{B}), (\mathrm{A}, \mathrm{D}), (\mathrm{D}, \mathrm{F}), (\mathrm{F}, \mathrm{G}), (\mathrm{G}, \mathrm{E}), (\mathrm{F}, \mathrm{C})$
D. $(\mathrm{A}, \mathrm{D}), (\mathrm{A}, \mathrm{B}), (\mathrm{D}, \mathrm{F}), (\mathrm{F}, \mathrm{C}), (\mathrm{F}, \mathrm{G}), (\mathrm{G}, \mathrm{E})$

gateit-2004 algorithms graph-algorithms normal prims-algorithm

Answer key

1.25.2 Prims Algorithm: GATE IT 2008 | Question: 45

For the undirected, weighted graph given below, which of the following sequences of edges represents a correct execution of Prim's algorithm to construct a Minimum Spanning Tree?



A. $(\mathrm{a}, \mathrm{b}), (\mathrm{d}, \mathrm{f}), (\mathrm{f}, \mathrm{c}), (\mathrm{g}, \mathrm{i}), (\mathrm{d}, \mathrm{a}), (\mathrm{g}, \mathrm{h}), (\mathrm{c}, \mathrm{e}), (\mathrm{f}, \mathrm{h})$
B. $(\mathrm{c}, \mathrm{e}), (\mathrm{c}, \mathrm{f}), (\mathrm{f}, \mathrm{d}), (\mathrm{d}, \mathrm{a}), (\mathrm{a}, \mathrm{b}), (\mathrm{g}, \mathrm{h}), (\mathrm{h}, \mathrm{f}), (\mathrm{g}, \mathrm{i})$
C. $(\mathrm{d}, \mathrm{f}), (\mathrm{f}, \mathrm{c}), (\mathrm{d}, \mathrm{a}), (\mathrm{a}, \mathrm{b}), (\mathrm{c}, \mathrm{e}), (\mathrm{f}, \mathrm{h}), (\mathrm{g}, \mathrm{h}), (\mathrm{g}, \mathrm{i})$
D. $(\mathrm{h}, \mathrm{g}), (\mathrm{g}, \mathrm{i}), (\mathrm{h}, \mathrm{f}), (\mathrm{f}, \mathrm{c}), (\mathrm{f}, \mathrm{d}), (\mathrm{d}, \mathrm{a}), (\mathrm{a}, \mathrm{b}), (\mathrm{c}, \mathrm{e})$

gateit-2008 algorithms graph-algorithms minimum-spanning-tree normal prims-algorithm

Answer key

1.26 Quick Sort (14)

1.26.1 Quick Sort: GATE CSE 1987 | Question: 1-xviii

Let P be a quicksort program to sort numbers in ascending order. Let t_1 and t_2 be the time taken by the program for the inputs $[1\ 2\ 3\ 4]$ and $[5\ 4\ 3\ 2\ 1]$, respectively. Which of the following holds?

A. $t_1 = t_2$
B. $t_1 > t_2$
C. $t_1 < t_2$
D. $t_1 = t_2 + 5 \log 5$

gate1987 algorithms sorting quick-sort

Answer key

1.26.2 Quick Sort: GATE CSE 1989 | Question: 9

An input files has 10 records with keys as given below:

$25 \quad 7 \quad 34 \quad 2 \quad 70 \quad 9 \quad 61 \quad 16 \quad 49 \quad 19$

This is to be sorted in non-decreasing order.

i. Sort the input file using QUICKSORT by correctly positioning the first element of the file/subfile. Show the subfiles obtained at all intermediate steps. Use square brackets to demarcate subfiles.
ii. Sort the input file using 2-way- MERGESORT showing all major intermediate steps. Use square brackets to demarcate subfiles.

gate1989 descriptive algorithms sorting quick-sort

Answer key

1.26.3 Quick Sort: GATE CSE 1992 | Question: 03,iv

Assume that the last element of the set is used as partition element in Quicksort. If n distinct elements from the set $[1 \dots n]$ are to be sorted, give an input for which Quicksort takes maximum time.

gate1992 algorithms sorting easy quick-sort descriptive

Answer key

1.26.4 Quick Sort: GATE CSE 1996 | Question: 2.15

Quick-sort is run on two inputs shown below to sort in ascending order taking first element as pivot

i. $1, 2, 3, \dots n$
ii. $n, n-1, n-2, \dots, 2, 1$

Let C_1 and C_2 be the number of comparisons made for the inputs (i) and (ii) respectively. Then,

A. $C_1 < C_2$
B. $C_1 > C_2$
C. $C_1 = C_2$
D. we cannot say anything for arbitrary n

gate1996 algorithms sorting normal quick-sort

Answer key

1.26.5 Quick Sort: GATE CSE 2001 | Question: 1.14

Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst case complexity of sorting n numbers using Randomized quicksort?

A. $O(n)$
B. $O(n \log n)$
C. $O(n^2)$
D. $O(n!)$

gatecse-2001 algorithms sorting time-complexity easy quick-sort

Answer key

1.26.6 Quick Sort: GATE CSE 2006 | Question: 52

The median of n elements can be found in $O(n)$ time. Which one of the following is correct about the complexity of quick sort, in which median is selected as pivot?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n^2)$
D. $\Theta(n^3)$

gatecse-2006 algorithms sorting easy quick-sort

Answer key

1.26.7 Quick Sort: GATE CSE 2008 | Question: 43

Consider the Quicksort algorithm. Suppose there is a procedure for finding a pivot element which splits the list into two sub-lists each of which contains at least one-fifth of the elements. Let $T(n)$ be the number of comparisons required to sort n elements. Then

A. $T(n) \leq 2T(n/5) + n$
B. $T(n) \leq T(n/5) + T(4n/5) + n$
C. $T(n) \leq 2T(4n/5) + n$
D. $T(n) \leq 2T(n/2) + n$

gatecse-2008 algorithms sorting easy quick-sort

Answer key

1.26.8 Quick Sort: GATE CSE 2009 | Question: 39

In quick-sort, for sorting n elements, the $(n/4)^{th}$ smallest element is selected as pivot using an $O(n)$ time algorithm. What is the worst case time complexity of the quick sort?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n^2)$
D. $\Theta(n^2 \log n)$

gatecse-2009 algorithms sorting normal quick-sort

Answer key

1.26.9 Quick Sort: GATE CSE 2014 Set 1 | Question: 14

Let P be quicksort program to sort numbers in ascending order using the first element as the pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $[1\ 2\ 3\ 4\ 5]$ and $[4\ 1\ 5\ 3\ 2]$ respectively. Which one of the following holds?

A. $t_1 = 5$
B. $t_1 < t_2$
C. $t_1 > t_2$
D. $t_1 = t_2$

gatecse-2014-set1 algorithms sorting easy quick-sort

Answer key

1.26.10 Quick Sort: GATE CSE 2014 Set 3 | Question: 14

You have an array of n elements. Suppose you implement quicksort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is

A. $O(n^2)$
B. $O(n \log n)$
C. $\Theta(n \log n)$
D. $O(n^3)$

gatecse-2014-set3 algorithms sorting easy quick-sort

Answer key

1.26.11 Quick Sort: GATE CSE 2015 Set 1 | Question: 2

Which one of the following is the recurrence equation for the worst case time complexity of the quick sort algorithm for sorting $n\ (\geq 2)$ numbers? In the recurrence equations given in the options below, c is a constant.

A. $T(n) = 2T(n/2) + cn$
B. $T(n) = T(n-1) + T(1) + cn$
C. $T(n) = 2T(n-1) + cn$
D. $T(n) = T(n/2) + cn$

gatecse-2015-set1 algorithms recurrence-relation sorting easy quick-sort

Answer key

1.26.12 Quick Sort: GATE CSE 2015 Set 2 | Question: 45

Suppose you are provided with the following function declaration in the C programming language.

```
int partition(int a[], int n);
```

The function treats the first element of $a[\,]$ as a pivot and rearranges the array so that all elements less than or equal to the pivot is in the left part of the array, and all elements greater than the pivot is in the right part. In addition, it moves the pivot so that the pivot is the last element of the left part. The return value is the number of elements in the left part.

The following partially given function in the C programming language is used to find the k^{th} smallest element in an array $a[\,]$ of size n using the partition function. We assume $k \leq n$.

```
int kth_smallest (int a[], int n, int k)
{
   int left_end = partition (a, n);
   if (left_end+1==k) {
      return a[left_end];
   }
   if (left_end+1 > k) {
      return kth_smallest (____________);
   } else {
      return kth_smallest (____________);
   }
}
```

The missing arguments lists are respectively

A. $(a,$ left_end$, k)$ and $(a+$left_end$+1, n-$left_end$-1, k-$left_end$-1)$
B. $(a,$ left_end$, k)$ and $(a, n-$left_end$-1, k-$left_end$-1)$
C. $(a+$left_end$+1, n-$left_end$-1, k-$left_end$-1)$ and $(a,$
D. $(a, n-$left_end$-1, k-$left_end$-1)$ and $(a,$left_end$, k)$

left_end, k)

gatecse-2015-set2 algorithms normal sorting quick-sort

Answer key

1.26.13 Quick Sort: GATE CSE 2019 | Question: 20

An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is ________

gatecse-2019 numerical-answers algorithms quick-sort probability one-mark

Answer key

1.26.14 Quick Sort: GATE DS&AI 2024 | Question: 20

Consider sorting the following array of integers in ascending order using an inplace Quicksort algorithm that uses the last element as the pivot.

60	70	80	90	100

The minimum number of swaps performed during this Quicksort is ________.

gate-ds-ai-2024 numerical-answers algorithms quick-sort one-mark

Answer key

1.27 Recurrence Relation (35)

1.27.1 Recurrence Relation: GATE CSE 1987 | Question: 10a

Solve the recurrence equations:

- $T(n) = T(n-1) + n$
- $T(1) = 1$

gate1987 algorithms recurrence-relation descriptive

Answer key

1.27.2 Recurrence Relation: GATE CSE 1988 | Question: 13iv

Solve the recurrence equations:

- $T(n) = T(\frac{n}{2}) + 1$
- $T(1) = 1$

gate1988 descriptive algorithms recurrence-relation

Answer key

1.27.3 Recurrence Relation: GATE CSE 1989 | Question: 13b

Find a solution to the following recurrence equation:

- $T(n) = \sqrt{n} + T\left(\frac{n}{2}\right)$
- $T(1) = 1$

gate1989 descriptive algorithms recurrence-relation

Answer key

1.27.4 Recurrence Relation: GATE CSE 1990 | Question: 17a

Express $T(n)$ in terms of the harmonic number $H_n = \sum_{i=1}^{n} \frac{1}{i}, \quad n \geq 1,$ where $T(n)$ satisfies the recurrence relation,

$T(n) = \frac{n+1}{n} T(n-1) + 1$, for $n \geq \sum$ and $T(1) = 1$

What is the asymptotic behaviour of $T(n)$ as a function of n ?

gate1990 descriptive algorithms recurrence-relation

Answer key

1.27.5 Recurrence Relation: GATE CSE 1992 | Question: 07a

Consider the function $F(n)$ for which the pseudocode is given below :

```
Function F(n)
begin
F1 ← 1
if(n=1) then F ← 3
 else
   For i = 1 to n do
       begin
         C ← 0
        For j = 1 to n – 1 do
        begin C ← C + 1 end
        F1 = F1 * C
      end
 F = F1
end
```

[n is a positive integer greater than zero]

A. Derive a recurrence relation for $F(n)$.

gate1992 algorithms recurrence-relation descriptive

Answer key

1.27.6 Recurrence Relation: GATE CSE 1992 | Question: 07b

Consider the function $F(n)$ for which the pseudocode is given below :

```
Function F(n)
begin
F1 ← 1
if(n=1) then F ← 3
 else
   For i = 1 to n do
       begin
         C ← 0
        For j = 1 to n – 1 do
        begin C ← C + 1 end
        F1 = F1 * C
      end
 F = F1
end
```

[n is a positive integer greater than zero]

B. Solve the recurrence relation for a closed form solution of $F(n)$.

gate1992 algorithms recurrence-relation descriptive

Answer key

1.27.7 Recurrence Relation: GATE CSE 1993 | Question: 15

Consider the recursive algorithm given below:

```
procedure bubblesort (n);
var i,j: index; temp : item;
begin
   for i:=1 to n-1 do
      if A[i] > A[i+1] then
      begin
         temp := A[i];
         A[i] := A[i+1];
         A[i+1] := temp;
      end;
   bubblesort (n-1)
end
```

Let a_n be the number of times the 'if…then…' statement gets executed when the algorithm is run with value n. Set up the recurrence relation by defining a_n in terms of a_{n-1}. Solve for a_n.

gate1993 algorithms recurrence-relation normal descriptive

Answer key

1.27.8 Recurrence Relation: GATE CSE 1994 | Question: 1.7, ISRO2017-14

The recurrence relation that arises in relation with the complexity of binary search is:

A. $T(n) = 2T\left(\frac{n}{2}\right) + k,$ k is a constant
B. $T(n) = T\left(\frac{n}{2}\right) + k,$ k is a constant
C. $T(n) = T\left(\frac{n}{2}\right) + \log n$
D. $T(n) = T\left(\frac{n}{2}\right) + n$

gate1994 algorithms recurrence-relation easy isro2017

Answer key

1.27.9 Recurrence Relation: GATE CSE 1996 | Question: 2.12

The recurrence relation

- $T(1) = 2$
- $T(n) = 3T(\frac{n}{4}) + n$

has the solution $T(n)$ equal to

A. $O(n)$
B. $O(\log n)$
C. $O\left(n^{\frac{3}{4}}\right)$
D. None of the above

gate1996 algorithms recurrence-relation normal

Answer key

1.27.10 Recurrence Relation: GATE CSE 1997 | Question: 15

Consider the following function.

```
Function F(n, m:integer):integer;
begin
   if (n<=0) or (m<=0) then F:=1
   else
     F:F(n-1, m) + F(n-1, m-1);
end;
```

Use the recurrence relation $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$ to answer the following questions. Assume that n, m are positive integers. Write only the answers without any explanation.

a. What is the value of $F(n, 2)$?
b. What is the value of $F(n, m)$?
c. How many recursive calls are made to the function F, including the original call, when evaluating $F(n, m)$.

gate1997 algorithms recurrence-relation descriptive

Answer key☟

1.27.11 Recurrence Relation: GATE CSE 1997 | Question: 4.6

Let $T(n)$ be the function defined by $T(1) = 1,\ T(n) = 2T(\lfloor \frac{n}{2} \rfloor) + \sqrt{n}$ for $n \geq 2$.
Which of the following statements is true?

A. $T(n) = O\sqrt{n}$
B. $T(n) = O(n)$
C. $T(n) = O(\log n)$
D. None of the above

gate1997 algorithms recurrence-relation normal

Answer key☟

1.27.12 Recurrence Relation: GATE CSE 1998 | Question: 6a

Solve the following recurrence relation

$$x_n = 2x_{n-1} - 1, n > 1$$

$$x_1 = 2$$

gate1998 algorithms recurrence-relation descriptive

Answer key☟

1.27.13 Recurrence Relation: GATE CSE 2002 | Question: 1.3

The solution to the recurrence equation $T(2^k) = 3T(2^{k-1}) + 1, T(1) = 1$ is

A. 2^k
B. $\frac{(3^{k+1}-1)}{2}$
C. $3^{\log_2 k}$
D. $2^{\log_3 k}$

gatecse-2002 algorithms recurrence-relation normal

Answer key☟

1.27.14 Recurrence Relation: GATE CSE 2002 | Question: 2.11

The running time of the following algorithm

Procedure $A(n)$

If $n \leqslant 2$ return (1) else return $(A(\lceil \sqrt{n} \rceil))$;

is best described by

A. $O(n)$
B. $O(\log n)$
C. $O(\log \log n)$
D. $O(1)$

gatecse-2002 algorithms recurrence-relation normal

Answer key☟

1.27.15 Recurrence Relation: GATE CSE 2003 | Question: 35

Consider the following recurrence relation

$T(1) = 1$

$T(n+1) = T(n) + \lfloor \sqrt{n+1} \rfloor$ for all $n \geq 1$

The value of $T(m^2)$ for $m \geq 1$ is

A. $\frac{m}{6}(21m - 39) + 4$
B. $\frac{m}{6}(4m^2 - 3m + 5)$
C. $\frac{m}{2}(3m^{2.5} - 11m + 20) - 5$
D. $\frac{m}{6}(5m^3 - 34m^2 + 137m - 104) + \frac{5}{6}$

gatecse-2003 algorithms time-complexity recurrence-relation difficult

Answer key☟

1.27.16 Recurrence Relation: GATE CSE 2004 | Question: 83, ISRO2015-40

The time complexity of the following C function is (assume $n > 0$)

```
int recursive (int n) {
   if(n == 1)
      return (1);
   else
      return (recursive (n-1) + recursive (n-1));
}
```

A. $O(n)$ B. $O(n \log n)$ C. $O(n^2)$ D. $O(2^n)$

gatecse-2004 algorithms recurrence-relation time-complexity normal isro2015

Answer key

1.27.17 Recurrence Relation: GATE CSE 2004 | Question: 84

The recurrence equation

$T(1) = 1$

$T(n) = 2T(n-1) + n, n \geq 2$

evaluates to

A. $2^{n+1} - n - 2$ B. $2^n - n$ C. $2^{n+1} - 2n - 2$ D. $2^n + n$

gatecse-2004 algorithms recurrence-relation normal

Answer key

1.27.18 Recurrence Relation: GATE CSE 2006 | Question: 51, ISRO2016-34

Consider the following recurrence:

$$T(n) = 2T\left(\sqrt{n}\right) + 1, T(1) = 1$$

Which one of the following is true?

A. $T(n) = \Theta(\log \log n)$
B. $T(n) = \Theta(\log n)$
C. $T(n) = \Theta(\sqrt{n})$
D. $T(n) = \Theta(n)$

algorithms recurrence-relation isro2016 gatecse-2006

Answer key

1.27.19 Recurrence Relation: GATE CSE 2008 | Question: 78

Let x_n denote the number of binary strings of length n that contain no consecutive 0s.

Which of the following recurrences does x_n satisfy?

A. $x_n = 2x_{n-1}$
B. $x_n = x_{\lfloor n/2 \rfloor} + 1$
C. $x_n = x_{\lfloor n/2 \rfloor} + n$
D. $x_n = x_{n-1} + x_{n-2}$

gatecse-2008 algorithms recurrence-relation normal

Answer key

1.27.20 Recurrence Relation: GATE CSE 2008 | Question: 79

Let x_n denote the number of binary strings of length n that contain no consecutive 0s.

The value of x_5 is

A. 5 B. 7 C. 8 D. 16

gatecse-2008 algorithms recurrence-relation normal

Answer key

1.27.21 Recurrence Relation: GATE CSE 2009 | Question: 35

The running time of an algorithm is represented by the following recurrence relation:

$$T(n) = \begin{cases} n & n \leq 3 \\ T(\frac{n}{3}) + cn & \text{otherwise} \end{cases}$$

Which one of the following represents the time complexity of the algorithm?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n^2)$
D. $\Theta(n^2 \log n)$

gatecse-2009 algorithms recurrence-relation time-complexity normal

Answer key

1.27.22 Recurrence Relation: GATE CSE 2012 | Question: 16

The recurrence relation capturing the optimal execution time of the $Towers\ of\ Hanoi$ problem with n discs is

A. $T(n) = 2T(n-2) + 2$
B. $T(n) = 2T(n-1) + n$
C. $T(n) = 2T(n/2) + 1$
D. $T(n) = 2T(n-1) + 1$

gatecse-2012 algorithms easy recurrence-relation

Answer key

1.27.23 Recurrence Relation: GATE CSE 2014 Set 2 | Question: 13

Which one of the following correctly determines the solution of the recurrence relation with $T(1) = 1$?

$$T(n) = 2T\left(\frac{n}{2}\right) + \log n$$

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n^2)$
D. $\Theta(\log n)$

gatecse-2014-set2 algorithms recurrence-relation normal

Answer key

1.27.24 Recurrence Relation: GATE CSE 2015 Set 1 | Question: 49

Let a$_n$ represent the number of bit strings of length n containing two consecutive 1s. What is the recurrence relation for a_n?

A. $a_{n-2} + a_{n-1} + 2^{n-2}$
B. $a_{n-2} + 2a_{n-1} + 2^{n-2}$
C. $2a_{n-2} + a_{n-1} + 2^{n-2}$
D. $2a_{n-2} + 2a_{n-1} + 2^{n-2}$

gatecse-2015-set1 algorithms recurrence-relation normal

Answer key

1.27.25 Recurrence Relation: GATE CSE 2015 Set 3 | Question: 39

Consider the following recursive C function.

```
void get(int n)
{
   if (n<1) return;
   get (n-1);
   get (n-3);
   printf("%d", n);
}
```

If $get(6)$ function is being called in $main()$ then how many times will the $get()$ function be invoked before returning to the $main()$?

A. 15
B. 25
C. 35
D. 45

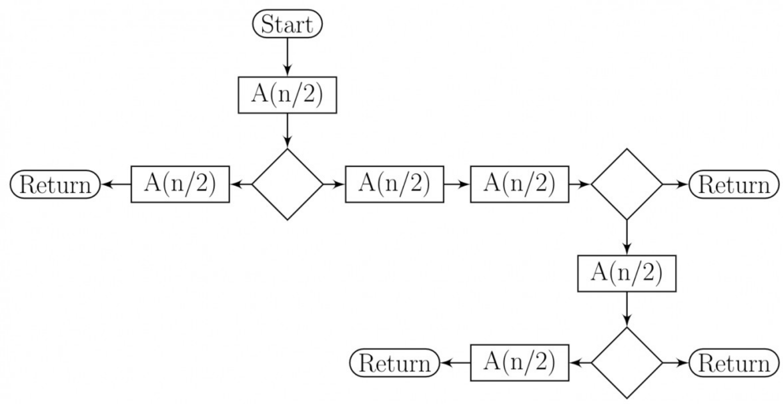
gatecse-2015-set3 algorithms recurrence-relation normal

Answer key

1.27.26 Recurrence Relation: GATE CSE 2016 Set 2 | Question: 39

The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^{\alpha})$, then the least possible value (accurate up to two decimal positions) of α is ________.

Flow chart for Recursive Function $A(n)$.



gatecse-2016-set2 algorithms time-complexity recurrence-relation normal numerical-answers

Answer key

1.27.27 Recurrence Relation: GATE CSE 2017 Set 2 | Question: 30

Consider the recurrence function

$$T(n) = \begin{cases} 2T(\sqrt{n}) + 1, & n > 2 \\ 2, & 0 < n \leq 2 \end{cases}$$

Then $T(n)$ in terms of Θ notation is

A. $\Theta(\log \log n)$
B. $\Theta(\log n)$
C. $\Theta(\sqrt{n})$
D. $\Theta(n)$

gatecse-2017-set2 algorithms recurrence-relation

Answer key

1.27.28 Recurrence Relation: GATE CSE 2020 | Question: 2

For parameters a and b, both of which are $\omega(1)$, $T(n) = T(n^{1/a}) + 1$, and $T(b) = 1$. Then $T(n)$ is

A. $\Theta(\log_a \log_b n)$
B. $\Theta(\log_{ab} n)$
C. $\Theta(\log_b \log_a n)$
D. $\Theta(\log_2 \log_2 n)$

gatecse-2020 algorithms recurrence-relation one-mark

Answer key

1.27.29 Recurrence Relation: GATE CSE 2021 Set 1 | Question: 30

Consider the following recurrence relation.

$$T(n) = \begin{cases} T(n/2) + T(2n/5) + 7n & \text{if } n > 0 \\ 1 & \text{if } n = 0 \end{cases}$$

Which one of the following options is correct?

A. $T(n) = \Theta(n^{5/2})$
B. $T(n) = \Theta(n \log n)$
C. $T(n) = \Theta(n)$
D. $T(n) = \Theta((\log n)^{5/2})$

gatecse-2021-set1 algorithms recurrence-relation time-complexity two-marks

Answer key

1.27.30 Recurrence Relation: GATE CSE 2021 Set 2 | Question: 39

For constants $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers:

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

Which one of the following options is correct about the recurrence $T(n)$?

A. If $f(n)$ is $n \log_2(n)$, then $T(n)$ is $\Theta(n \log_2(n))$
B. If $f(n)$ is $\dfrac{n}{\log_2(n)}$, then $T(n)$ is $\Theta(\log_2(n))$
C. If $f(n)$ is $O(n^{\log_b(a)-\epsilon})$ for some $\epsilon > 0$, then $T(n)$ is $\Theta(n^{\log_b(a)})$
D. If $f(n)$ is $\Theta(n^{\log_b(a)})$, then $T(n)$ is $\Theta(n^{\log_b(a)})$

gatecse-2021-set2 algorithms recurrence-relation two-marks

Answer key☟

1.27.31 Recurrence Relation: GATE CSE 2024 | Set 1 | Question: 32

Consider the following recurrence relation:

$$T(n) = \begin{cases} \sqrt{n}T(\sqrt{n}) + n \text{ for } n \geq 1, \\ \quad 1 \quad \text{for } n = 1 \end{cases}$$

Which one of the following options is CORRECT?

A. $T(n) = \Theta(n \log \log n)$
B. $T(n) = \Theta(n \log n)$
C. $T(n) = \Theta\left(n^2 \log n\right)$
D. $T(n) = \Theta\left(n^2 \log \log n\right)$

gatecse2024-set1 algorithms recurrence-relation two-marks

Answer key☟

1.27.32 Recurrence Relation: GATE CSE 2025 | Set 1 | Question: 10

Consider the following recurrence relation:

$$T(n) = 2T(n-1) + n2^n \text{ for } n > 0, \quad T(0) = 1$$

Which ONE of the following options is CORRECT?

A. $T(n) = \Theta\left(n^2 2^n\right)$
B. $T(n) = \Theta\left(n2^n\right)$
C. $T(n) = \Theta\left((\log n)^2 2^n\right)$
D. $T(n) = \Theta\left(4^n\right)$

gatecse2025-set1 algorithms time-complexity recurrence-relation one-mark

Answer key☟

1.27.33 Recurrence Relation: GATE IT 2004 | Question: 57

Consider a list of recursive algorithms and a list of recurrence relations as shown below. Each recurrence relation corresponds to exactly one algorithm and is used to derive the time complexity of the algorithm.

	Recursive Algorithm		**Recurrence Relation**
P	Binary search	l.	$T(n) = T(n-k) + T(k) + cn$
Q.	Merge sort	ll.	$T(n) = 2T(n-1) + 1$
R.	Quick sort	lll.	$T(n) = 2T(n/2) + cn$
S.	Tower of Hanoi	lV.	$T(n) = T(n/2) + 1$

Which of the following is the correct match between the algorithms and their recurrence relations?

A. P-II, Q-III, R-IV, S-I
B. P-IV, Q-III, R-I, S-II
C. P-III, Q-II, R-IV, S-I
D. P-IV, Q-II, R-I, S-III

gateit-2004 algorithms recurrence-relation normal match-the-following

Answer key☟

1.27.34 Recurrence Relation: GATE IT 2005 | Question: 51

Let $T(n)$ be a function defined by the recurrence

$T(n) = 2T(n/2) + \sqrt{n}$ for $n \geq 2$ and
$T(1) = 1$

Which of the following statements is **TRUE**?

A. $T(n) = \Theta(\log n)$
B. $T(n) = \Theta(\sqrt{n})$
C. $T(n) = \Theta(n)$
D. $T(n) = \Theta(n \log n)$

gateit-2005 algorithms recurrence-relation easy

Answer key

1.27.35 Recurrence Relation: GATE IT 2008 | Question: 44

When $n = 2^{2k}$ for some $k \geqslant 0$, the recurrence relation

$T(n) = \sqrt(2)T(n/2) + \sqrt{n}, T(1) = 1$

evaluates to :

A. $\sqrt(n)(\log n + 1)$
B. $\sqrt(n) \log n$
C. $\sqrt(n) \log \sqrt(n)$
D. $n \log \sqrt{n}$

gateit-2008 algorithms recurrence-relation normal

Answer key

1.28 Recursion (4)

1.28.1 Recursion: GATE CSE 1995 | Question: 2.9

A language with string manipulation facilities uses the following operations

```
head(s): first character of a string
tail(s): all but exclude the first character of a string
```

```
concat(s1, s2): s1s2
```

For the string "$acbc$" what will be the output of

```
concat(head(s), head(tail(tail(s))))
```

A. ac
B. bc
C. ab
D. cc

gate1995 algorithms normal recursion

Answer key

1.28.2 Recursion: GATE CSE 2007 | Question: 44

In the following C function, let $n \geq m$.

```
int gcd(n,m)  {
   if (n%m == 0) return m;
   n = n%m;
   return gcd(m,n);
}
```

How many recursive calls are made by this function?

A. $\Theta(\log_2 n)$
B. $\Omega(n)$
C. $\Theta(\log_2 \log_2 n)$
D. $\Theta(\sqrt{n})$

gatecse-2007 algorithms recursion time-complexity normal

Answer key

1.28.3 Recursion: GATE CSE 2018 | Question: 45

Consider the following program written in pseudo-code. Assume that x and y are integers.

```
Count (x, y) {
```

```
    if (y !=1 ) {
       if (x !=1) {
          print("*");
          Count (x/2, y);
       }
       else {
          y=y-1;
          Count (1024, y);
       }
    }
}
```

The number of times that the $print$ statement is executed by the call $Count(1024, 1024)$ is _____

gatecse-2018 numerical-answers algorithms recursion two-marks

Answer key

1.28.4 Recursion: GATE CSE 2021 Set 2 | Question: 49

Consider the following ANSI C program

```
#include <stdio.h>
int foo(int x, int y, int q)
   {
      if ((x<=0) && (y<=0))
      return q;
      if (x<=0)
      return foo(x, y-q, q);
      if (y<=0)
      return foo(x-q, y, q);
      return foo(x, y-q, q) + foo(x-q, y, q);
   }
int main( )
{
   int r = foo(15, 15, 10);
   printf(“%d”, r);
   return 0;
}
```

The output of the program upon execution is _________

gatecse-2021-set2 algorithms recursion output numerical-answers two-marks

Answer key

1.29 Searching (7)

1.29.1 Searching: GATE CSE 1996 | Question: 18

Consider the following program that attempts to locate an element x in an array $a[]$ using binary search. Assume $N > 1$. The program is erroneous. Under what conditions does the program fail?

```
var i,j,k: integer; x: integer;
   a: array; [1..N] of integer;
begin i:= 1; j:= n;
repeat
   k:(i+j) div 2;
   if a[k] < x then i:= k
   else j:= k
until (a[k] = x) or (i >= j);

if (a[k] = x) then
   writeln ('x is in the array')
else
   writeln ('x is not in the array')
end;
```

gate1996 algorithms searching normal descriptive

Answer key

1.29.2 Searching: GATE CSE 1996 | Question: 2.13, ISRO2016-28

The average number of key comparisons required for a successful search for sequential search on n items is

A. $\frac{n}{2}$ B. $\frac{n-1}{2}$ C. $\frac{n+1}{2}$ D. None of the above

gate1996 algorithms easy isro2016 searching

Answer key

1.29.3 Searching: GATE CSE 2002 | Question: 2.10

Consider the following algorithm for searching for a given number x in an unsorted array $A[1..n]$ having n distinct values:

1. Choose an i at random from $1..n$
2. If $A[i] = x$, then Stop else Goto 1;

Assuming that x is present in A, what is the expected number of comparisons made by the algorithm before it terminates?

A. n B. $n-1$ C. $2n$ D. $\frac{n}{2}$

gatecse-2002 searching normal

Answer key

1.29.4 Searching: GATE CSE 2008 | Question: 84

Consider the following C program that attempts to locate an element x in an array $Y[\,]$ using binary search. The program is erroneous.

```
f (int Y[10] , int x) {
  int i, j, k;
  i= 0; j = 9;
  do {
     k = (i+ j) / 2;
     if( Y[k] < x) i = k;else j = k;
     } while (Y[k] != x) && (i < j)) ;
  if(Y[k] == x) printf(" x is in the array ") ;
  else printf(" x is not in the array ") ;
}
```

On which of the following contents of Y and x does the program fail?

A. Y is $[1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10]$ and $x < 10$
B. Y is $[1\ 3\ 5\ 7\ 9\ 11\ 13\ 15\ 17\ 19]$ and $x < 1$
C. Y is $[2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2]$ and $x > 2$
D. Y is $[2\ 4\ 6\ 8\ 10\ 12\ 14\ 16\ 18\ 20]$ and $2 < x < 20$ and x is even

gatecse-2008 algorithms searching normal

Answer key

1.29.5 Searching: GATE CSE 2008 | Question: 85

Consider the following C program that attempts to locate an element x in an array $Y[\,]$ using binary search. The program is erroneous.

```
f (int Y[10] , int x) {
  int i, j, k;
  i= 0; j = 9;
  do {
     k = (i + j) / 2;
     if( Y[k] < x) i = k;else j = k;
     } while (Y[k] != x) && (i < j)) ;
  if(Y[k] == x) printf(" x is in the array ") ;
  else printf(" x is not in the array ") ;
}
```

The correction needed in the program to make it work properly is

A. Change line 6 to: if $(Y[k] < x) i = k + 1$; else $j = k - 1$;

B. Change line 6 to: if $(Y[k] < x) i = k - 1$; else $j = k + 1$;
C. Change line 6 to: if $(Y[k] < x) i = k$; else $j = k$;
D. Change line 7 to: } while $((Y[k] == x) \&\& (i < j))$;

gatecse-2008 algorithms searching normal

Answer key

1.29.6 Searching: GATE CSE 2017 Set 1 | Question: 48

Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A. The *worst case* number of probes performed by an *optimal* algorithm is ____________.

gatecse-2017-set1 algorithms normal numerical-answers searching

Answer key

1.29.7 Searching: GATE CSE 2025 | Set 2 | Question: 19

Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?

A. A DFS tree of G is a Shortest Path tree of G.
B. Every non-tree edge of G with respect to a DFS tree is a forward/back edge.
C. If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.
D. Both BFS and DFS can be used to find the connected components of G.

gatecse2025-set2 algorithms searching breadth-first-search depth-first-search multiple-selects one-mark

Answer key

1.30 Shortest Path (9)

1.30.1 Shortest Path: GATE CSE 2002 | Question: 12

Fill in the blanks in the following template of an algorithm to compute all pairs shortest path lengths in a directed graph G with $n * n$ adjacency matrix A. $A[i, j]$ equals 1 if there is an edge in G from i to j, and 0 otherwise. Your aim in filling in the blanks is to ensure that the algorithm is correct.

```
INITIALIZATION: For i = 1 ... n
   {For j = 1 ... n
      { if a[i,j] = 0 then P[i,j] =_______ else P[i,j] =_______;}
   }

ALGORITHM: For i = 1 ... n
   {For j = 1 ... n
      {For k = 1 ... n
         {P[__,__] = min{_______,______}; }
      }
   }
```

a. Copy the complete line containing the blanks in the Initialization step and fill in the blanks.
b. Copy the complete line containing the blanks in the Algorithm step and fill in the blanks.
c. Fill in the blank: The running time of the Algorithm is O(___).

gatecse-2002 algorithms graph-algorithms time-complexity normal descriptive shortest-path

Answer key

1.30.2 Shortest Path: GATE CSE 2003 | Question: 67

Let $G = (V, E)$ be an undirected graph with a subgraph $G_1 = (V_1, E_1)$. Weights are assigned to edges of G as follows.

$$w(e) = \begin{cases} 0, & \text{if } e \in E_1 \\ 1, & \text{otherwise} \end{cases}$$

A single-source shortest path algorithm is executed on the weighted graph (V, E, w) with an arbitrary vertex v_1 of V_1 as the source. Which of the following can always be inferred from the path costs computed?

A. The number of edges in the shortest paths from v_1 to all vertices of G
B. G_1 is connected
C. V_1 forms a clique in G
D. G_1 is a tree

gatecse-2003 algorithms graph-algorithms normal shortest-path

Answer key

1.30.3 Shortest Path: GATE CSE 2007 | Question: 41

In an unweighted, undirected connected graph, the shortest path from a node S to every other node is computed most efficiently, in terms of *time complexity*, by

A. Dijkstra's algorithm starting from S.
B. Warshall's algorithm.
C. Performing a DFS starting from S.
D. Performing a BFS starting from S.

gatecse-2007 algorithms graph-algorithms easy shortest-path

Answer key

1.30.4 Shortest Path: GATE CSE 2020 | Question: 40

Let $G = (V, E)$ be a directed, weighted graph with weight function $w : E \to \mathbb{R}$. For some function $f : V \to \mathbb{R}$, for each edge $(u, v) \in E$, define $w'(u, v)$ as $w(u, v) + f(u) - f(v)$.

Which one of the options completes the following sentence so that it is TRUE?

"The shortest paths in G under w are shortest paths under w' too,_____________".

A. for every $f : V \to \mathbb{R}$
B. if and only if $\forall u \in V,\ f(u)$ is positive
C. if and only if $\forall u \in V,\ f(u)$ is negative
D. if and only if $f(u)$ is the distance from s to u in the graph obtained by adding a new vertex s to G and edges of zero weight from s to every vertex of G

gatecse-2020 algorithms graph-algorithms two-marks shortest-path

Answer key

1.30.5 Shortest Path: GATE CSE 2025 | Set 1 | Question: 33

Let $G(V, E)$ be an undirected and unweighted graph with 100 vertices. Let $d(u, v)$ denote the number of edges in a shortest path between vertices u and v in V. Let the maximum value of $d(u, v), u, v \in V$ such that $u \neq v$, be 30. Let T be any breadth-first-search tree of G. Which ONE of the given options is CORRECT for every such graph G ?

A. The height of T is exactly 15.
B. The height of T is exactly 30.
C. The height of T is at least 15.
D. The height of T is at least 30.

gatecse2025-set1 algorithms breadth-first-search shortest-path two-marks

Answer key

1.30.6 Shortest Path: GATE CSE 2025 | Set 1 | Question: 8

Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let $d_1(u, v)$ and $d_2(u, v)$ be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v ?

A. $d_1(u,v) = d_2(u,v)$
B. $d_1(u,v) \leq d_2(u,v)$

C. $d_1(u,v) \geq d_2(u,v)$

D. $d_1(u,v) \neq d_2(u,v)$

gatecse2025-set1 algorithms minimum-spanning-tree shortest-path one-mark

Answer key

1.30.7 Shortest Path: GATE CSE 2025 | Set 2 | Question: 27

Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge.

Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?

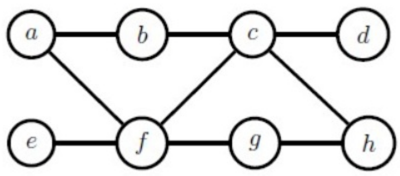
A. Every MST remains an MST, and every SP remains an SP.
B. MSTs need not remain MSTs, and every SP remains an SP.
C. Every MST remains an MST, and SPs need not remain SPs.
D. MSTs need not remain MSTs, and SPs need not remain SPs.

gatecse2025-set2 algorithms minimum-spanning-tree shortest-path two-marks

Answer key

1.30.8 Shortest Path: GATE DA 2025 | Question: 48

Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.



It is given that $a-b-c-d$ is a shortest path between a and d; $e-f-g-h$ is a shortest path between e and h; $a-f-c-h$ is a shortest path between a and h. Which of the following is/are NOT the edges of G ?

A. (b,d)

B. (b,g)

C. (b,h)

D. (e,g)

gateda-2025 algorithms shortest-path multiple-selects two-marks

1.30.9 Shortest Path: GATE IT 2007 | Question: 3, UGCNET-June2012-III: 34

Consider a weighted, undirected graph with positive edge weights and let uv be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements is always TRUE?

A. Weight $(u,v) \leq 12$

B. Weight $(u,v) = 12$

C. Weight $(u,v) \geq 12$

D. Weight $(u,v) > 12$

gateit-2007 algorithms graph-algorithms normal ugcnetcse-june2012-paper3 shortest-path

Answer key

1.31 Sorting (28)

1.31.1 Sorting: GATE CSE 1988 | Question: 1iii

Quicksort is ________ efficient than heapsort in the worst case.

gate1988 algorithms sorting fill-in-the-blanks easy

Answer key

1.31.2 Sorting: GATE CSE 1990 | Question: 3-v

The complexity of comparison based sorting algorithms is:

A. $\Theta(n \log n)$
B. $\Theta(n)$
C. $\Theta\left(n^2\right)$
D. $\Theta(n\sqrt{n})$

gate1990 normal algorithms sorting easy time-complexity multiple-selects

Answer key

1.31.3 Sorting: GATE CSE 1991 | Question: 01,vii

The minimum number of comparisons required to sort 5 elements is ______

gate1991 normal algorithms sorting numerical-answers

Answer key

1.31.4 Sorting: GATE CSE 1991 | Question: 13

Give an optimal algorithm in pseudo-code for sorting a sequence of n numbers which has only k distinct numbers (k is not known a Priori). Give a brief analysis for the time-complexity of your algorithm.

gate1991 sorting time-complexity algorithms difficult descriptive

Answer key

1.31.5 Sorting: GATE CSE 1992 | Question: 02,ix

Following algorithm(s) can be used to sort n in the range $[1 \dots n^3]$ in $O(n)$ time

a. Heap sort
b. Quick sort
c. Merge sort
d. Radix sort

gate1992 easy algorithms sorting multiple-selects

Answer key

1.31.6 Sorting: GATE CSE 1995 | Question: 12

Consider the following sequence of numbers:

$$92, 37, 52, 12, 11, 25$$

Use Bubble sort to arrange the sequence in ascending order. Give the sequence at the end of each of the first five passes.

gate1995 algorithms sorting easy descriptive bubble-sort

Answer key

1.31.7 Sorting: GATE CSE 1996 | Question: 14

A two dimensional array $A[1..n][1..n]$ of integers is partially sorted if $\forall i, j \in [1..n-1], A[i][j] < A[i][j+1] \text{ and } A[i][j] < A[i+1][j]$

a. The smallest item in the array is at $A[i][j]$ where $i =$ __ and $j =$ __.
b. The smallest item is deleted. Complete the following $O(n)$ procedure to insert item x (which is guaranteed to be smaller than any item in the last row or column) still keeping A partially sorted.

```
procedure insert (x: integer);
var i,j: integer;
begin
   i:=1; j:=1, A[i][j]:=x;
   while (x > __ or x > __) do
      if A[i+1][j] < A[i][j] ___ then begin
         A[i][j]:=A[i+1][j]; i:=i+1;
      end
      else begin
         _____
      end
   A[i][j]:= _____
end
```

gate1996 algorithms sorting normal descriptive

Answer key

1.31.8 Sorting: GATE CSE 1998 | Question: 1.22

Give the correct matching for the following pairs:

(A)	$O(\log n)$	(P)	Selection
(B)	$O(n)$	(Q)	Insertion sort
(C)	$O(n \log n)$	(R)	Binary search
(D)	$O(n^2)$	(S)	Merge sort

A. A-R B-P C-Q D-S

B. A-R B-P C-S D-Q

C. A-P B-R C-S D-Q

D. A-P B-S C-R D-Q

gate1998 algorithms sorting easy match-the-following

Answer key

1.31.9 Sorting: GATE CSE 1999 | Question: 1.12

A sorting technique is called stable if

A. it takes $O(n \log n)$ time
B. it maintains the relative order of occurrence of non-distinct elements
C. it uses divide and conquer paradigm
D. it takes $O(n)$ space

gate1999 algorithms sorting easy

Answer key

1.31.10 Sorting: GATE CSE 1999 | Question: 8

Let A be an $n \times n$ matrix such that the elements in each row and each column are arranged in ascending order. Draw a decision tree, which finds 1^{st}, 2^{nd} and 3^{rd} smallest elements in minimum number of comparisons.

gate1999 algorithms sorting normal descriptive

Answer key

1.31.11 Sorting: GATE CSE 2000 | Question: 17

An array contains four occurrences of 0, five occurrences of 1, and three occurrences of 2 in any order. The array is to be sorted using swap operations (elements that are swapped need to be adjacent).

a. What is the minimum number of swaps needed to sort such an array in the worst case?
b. Give an ordering of elements in the above array so that the minimum number of swaps needed to sort the array is maximum.

gatecse-2000 algorithms sorting normal descriptive

Answer key

1.31.12 Sorting: GATE CSE 2003 | Question: 61

In a permutation $a_1 \dots a_n$, of n distinct integers, an inversion is a pair (a_i, a_j) such that $i < j$ and $a_i > a_j$.

If all permutations are equally likely, what is the expected number of inversions in a randomly chosen permutation of $1 \dots n$?

A. $\frac{n(n-1)}{2}$

B. $\frac{n(n-1)}{4}$

C. $\frac{n(n+1)}{4}$

D. $2n[\log_2 n]$

gatecse-2003 algorithms sorting inversion normal

Answer key

1.31.13 Sorting: GATE CSE 2005 | Question: 39

Suppose there are $\lceil \log n \rceil$ sorted lists of $\lfloor n/\log n \rfloor$ elements each. The time complexity of producing a sorted list of all these elements is: (Hint:Use a heap data structure)

A. $O(n \log \log n)$
B. $\Theta(n \log n)$
C. $\Omega(n \log n)$
D. $\Omega\left(n^{3/2}\right)$

gatecse-2005 algorithms sorting normal

Answer key

1.31.14 Sorting: GATE CSE 2006 | Question: 14, ISRO2011-14

Which one of the following in place sorting algorithms needs the minimum number of swaps?

A. Quick sort
B. Insertion sort
C. Selection sort
D. Heap sort

gatecse-2006 algorithms sorting easy isro2011

Answer key

1.31.15 Sorting: GATE CSE 2007 | Question: 14

Which of the following sorting algorithms has the lowest worse-case complexity?

A. Merge sort
B. Bubble sort
C. Quick sort
D. Selection sort

gatecse-2007 algorithms sorting time-complexity easy

Answer key

1.31.16 Sorting: GATE CSE 2009 | Question: 11

What is the number of swaps required to sort n elements using selection sort, in the worst case?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n^2)$
D. $\Theta(n^2 \log n)$

gatecse-2009 algorithms sorting easy selection-sort

Answer key

1.31.17 Sorting: GATE CSE 2013 | Question: 30

The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

A. $\Theta(1)$
B. $\Theta(\sqrt{\log} n)$
C. $\Theta(\frac{\log n}{\log \log n})$
D. $\Theta(\log n)$

gatecse-2013 algorithms sorting normal heap-sort

Answer key

1.31.18 Sorting: GATE CSE 2013 | Question: 6

Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

A. $O(\log n)$
B. $O(n)$
C. $O(n \log n)$
D. $O(n^2)$

gatecse-2013 algorithms sorting easy selection-sort

Answer key

1.31.19 Sorting: GATE CSE 2014 Set 1 | Question: 39

The minimum number of comparisons required to find the minimum and the maximum of 100 numbers is ________ gatecse-2014-set1 algorithms numerical-answers normal maximum-minimum sorting

Answer key

1.31.20 Sorting: GATE CSE 2016 Set 1 | Question: 13

The worst case running times of *Insertion sort* , *Merge sort* and *Quick sort*, respectively are:

A. $\Theta(n \log n)$, $\Theta(n \log n)$ and $\Theta(n^2)$
B. $\Theta(n^2)$, $\Theta(n^2)$ and $\Theta(n \log n)$
C. $\Theta(n^2)$, $\Theta(n \log n)$ and $\Theta(n \log n)$
D. $\Theta(n^2)$, $\Theta(n \log n)$ and $\Theta(n^2)$

gatecse-2016-set1 algorithms sorting easy

Answer key

1.31.21 Sorting: GATE CSE 2016 Set 2 | Question: 13

Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in the ascending order, which of the following are **TRUE**?

I. Quicksort runs in $\Theta(n^2)$ time
II. Bubblesort runs in $\Theta(n^2)$ time
III. Mergesort runs in $\Theta(n)$ time
IV. Insertion sort runs in $\Theta(n)$ time

A. I and II only
B. I and III only
C. II and IV only
D. I and IV only

gatecse-2016-set2 algorithms sorting time-complexity normal ambiguous

Answer key

1.31.22 Sorting: GATE CSE 2021 Set 1 | Question: 9

Consider the following array.

23	32	45	69	72	73	89	97

Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?

A. Selection sort
B. Mergesort
C. Insertion sort
D. Quicksort using the last element as pivot

gatecse-2021-set1 algorithms sorting one-mark

Answer key

1.31.23 Sorting: GATE CSE 2024 | Set 1 | Question: 31

An array $[82, 101, 90, 11, 111, 75, 33, 131, 44, 93]$ is heapified. Which one of the following options represents the first three elements in the heapified array?

A. $82, 90, 101$
B. $82, 11, 93$
C. $131, 11, 93$
D. $131, 111, 90$

gatecse2024-set1 algorithms heap-sort sorting two-marks

Answer key

1.31.24 Sorting: GATE CSE 2024 | Set 2 | Question: 25

Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array $[2, 5, 3, 1, 4, 2, 6]$ is ___________.

gatecse2024-set2 numerical-answers algorithms sorting one-mark

Answer key

1.31.25 Sorting: GATE CSE 2025 | Set 2 | Question: 10

Consider an unorders list of N distinct integers.

What is the minimum number of element comaparisons required to find an integer in the list that is NOT the largest in the list?

A. 1 B. $N-1$ C. N D. $2N-1$

gatecse2025-set2 algorithms sorting one-mark

Answer key

1.31.26 Sorting: GATE DS&AI 2024 | Question: 35

Consider the following sorting algorithms:

i. Bubble sort
ii. Insertion sort
iii. Selection sort

Which ONE among the following choices of sorting algorithms sorts the numbers in the array $[4, 3, 2, 1, 5]$ in increasing order after exactly two passes over the array?

A. (i) only B. (iii) only C. (i) and (iii) only D. (ii) and (iii) only

gate-ds-ai-2024 algorithms sorting two-marks

Answer key

1.31.27 Sorting: GATE IT 2005 | Question: 59

Let a and b be two sorted arrays containing n integers each, in non-decreasing order. Let c be a sorted array containing $2n$ integers obtained by merging the two arrays a and b. Assuming the arrays are indexed starting from 0, consider the following four statements

I. $a[i] \geq b[i] \Rightarrow c[2i] \geq a[i]$
II. $a[i] \geq b[i] \Rightarrow c[2i] \geq b[i]$
III. $a[i] \geq b[i] \Rightarrow c[2i] \leq a[i]$
IV. $a[i] \geq b[i] \Rightarrow c[2i] \leq b[i]$

Which of the following is TRUE?

A. only I and II B. only I and IV C. only II and III D. only III and IV

gateit-2005 algorithms sorting normal

Answer key

1.31.28 Sorting: GATE IT 2008 | Question: 43

If we use Radix Sort to sort n integers in the range $\left(n^{k/2}, n^k\right]$, for some $k > 0$ which is independent of n, the time taken would be?

A. $\Theta(n)$ B. $\Theta(kn)$ C. $\Theta(n \log n)$ D. $\Theta(n^2)$

gateit-2008 algorithms sorting normal

Answer key

1.32 Space Complexity (1)

1.32.1 Space Complexity: GATE CSE 2005 | Question: 81a

```
double foo(int n)
```

```
{
   int i;
   double sum;
   if(n == 0)
   {
      return 1.0;
   }
   else
   {
      sum = 0.0;
      for(i = 0; i < n; i++)
      {
         sum += foo(i);
      }
      return sum;
   }
}
```

The space complexity of the above code is?

A. $O(1)$ B. $O(n)$ C. $O(n!)$ D. n^n

gatecse-2005 algorithms recursion normal space-complexity

Answer key

1.33 Strongly Connected Components (3)

1.33.1 Strongly Connected Components: GATE CSE 2008 | Question: 7

The most efficient algorithm for finding the number of connected components in an undirected graph on n vertices and m edges has time complexity

A. $\Theta(n)$ B. $\Theta(m)$ C. $\Theta(m+n)$ D. $\Theta(mn)$

gatecse-2008 algorithms graph-algorithms time-complexity normal strongly-connected-components

Answer key

1.33.2 Strongly Connected Components: GATE CSE 2018 | Question: 43

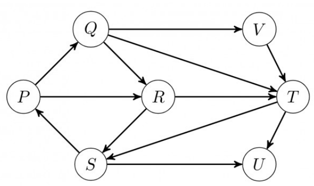
Let G be a graph with $100!$ vertices, with each vertex labelled by a distinct permutation of the numbers $1, 2, \ldots, 100$. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v. Let y denote the degree of a vertex in G, and z denote the number of connected components in G. Then, $y + 10z =$ ______.

gatecse-2018 algorithms graph-algorithms numerical-answers two-marks strongly-connected-components

Answer key

1.33.3 Strongly Connected Components: GATE IT 2006 | Question: 46

Which of the following is the correct decomposition of the directed graph given below into its strongly connected components?



A. $\{P, Q, R, S\}, \{T\}, \{U\}, \{V\}$
B. $\{P, Q, R, S, T, V\}, \{U\}$
C. $\{P, Q, S, T, V\}, \{R\}, \{U\}$
D. $\{P, Q, R, S, T, U, V\}$

gateit-2006 algorithms graph-algorithms normal strongly-connected-components

Answer key

1.34.1 Time Complexity: GATE CSE 1988 | Question: 6i

Given below is the sketch of a program that represents the path in a two-person game tree by the sequence of active procedure calls at any time. The program assumes that the payoffs are real number in a limited range; that the constant INF is larger than any positive payoff and its negation is smaller than any negative payoff and that there is a function "payoff" and that computes the payoff for any board that is a leaf. The type "boardtype" has been suitably declared to represent board positions. It is player-1's move if mode = MAX and player-2's move if mode=MIN. The type modetype = (MAX, MIN). The functions "min" and "max" find the minimum and maximum of two real numbers.

```
function search(B: boardtype; mode: modetype): real;
  var
    C:boardtype; {a child of board B}
    value:real;
  begin
    if B is a leaf then
      return (payoff(B))
    else
      begin
      if mode = MAX then value :=-INF
      else
        value:INF;
      for each child C of board B do
      if mode = MAX then
        value:=max (value, search (C, MIN))
      else
        value:=min(value, search(C, MAX))
      return(value)
      end
  end; (search)
```

Comment on the working principle of the above program. Suggest a possible mechanism for reducing the amount of search.

gate1988 normal descriptive algorithms time-complexity

Answer key

1.34.2 Time Complexity: GATE CSE 1989 | Question: 2-iii

Match the pairs in the following:

(A)	$O(\log n)$	(p)	Heapsort
(B)	$O(n)$	(q)	Depth-first search
(C)	$O(n \log n)$	(r)	Binary search
(D)	$O(n^2)$	(s)	Selection of the k^{th} smallest element in a set of n elements

gate1989 match-the-following algorithms time-complexity

Answer key

1.34.3 Time Complexity: GATE CSE 1993 | Question: 8.7

$\sum\limits_{1 \leq k \leq n} O(n)$, where $O(n)$ stands for order n is:

A. $O(n)$
B. $O(n^2)$
C. $O(n^3)$
D. $O(3n^2)$
E. $O(1.5n^2)$

gate1993 algorithms time-complexity easy

Answer key

1.34.4 Time Complexity: GATE CSE 1999 | Question: 1.13

Suppose we want to arrange the n numbers stored in any array such that all negative values occur before all positive ones. Minimum number of exchanges required in the worst case is

A. $n-1$ B. n C. $n+1$ D. None of the above

gate1999 algorithms time-complexity normal

Answer key

1.34.5 Time Complexity: GATE CSE 1999 | Question: 1.16

If n is a power of 2, then the minimum number of multiplications needed to compute a^n is

A. $\log_2 n$ B. $\sqrt{n}$ C. $n-1$ D. n

gate1999 algorithms time-complexity normal

Answer key

1.34.6 Time Complexity: GATE CSE 1999 | Question: 11a

Consider the following algorithms. Assume, procedure A and procedure B take $O(1)$ and $O(1/n)$ unit of time respectively. Derive the time complexity of the algorithm in O-notation.

```
algorithm what (n)
begin
    if n = 1 then call A
    else
        begin
            what (n-1);
            call B(n)
        end
end.
```

gate1999 algorithms time-complexity normal descriptive

Answer key

1.34.7 Time Complexity: GATE CSE 2000 | Question: 1.15

Let S be a sorted array of n integers. Let $T(n)$ denote the time taken for the most efficient algorithm to determined if there are two elements with sum less than 1000 in S. Which of the following statement is true?

A. $T(n)$ is $O(1)$
B. $n \leq T(n) \leq n \log_2 n$
C. $n \log_2 n \leq T(n) < \frac{n}{2}$
D. $T(n) = \binom{n}{2}$

gatecse-2000 easy algorithms time-complexity

Answer key

1.34.8 Time Complexity: GATE CSE 2003 | Question: 66

The cube root of a natural number n is defined as the largest natural number m such that $(m^3 \leq n)$. The complexity of computing the cube root of n (n is represented by binary notation) is

A. $O(n)$ but not $O(n^{0.5})$
B. $O(n^{0.5})$ but not $O((\log n)^k)$ for any constant $k > 0$
C. $O((\log n)^k)$ for some constant $k > 0$, but not $O((\log \log n)^m)$ for any constant $m > 0$
D. $O((\log \log n)^k)$ for some constant $k > 0.5$, but not $O((\log \log n)^{0.5})$

gatecse-2003 algorithms time-complexity normal

Answer key

1.34.9 Time Complexity: GATE CSE 2004 | Question: 39

Two matrices M_1 and M_2 are to be stored in arrays A and B respectively. Each array can be stored either in row-major or column-major order in contiguous memory locations. The time complexity of an algorithm to compute $M_1 \times M_2$ will be

A. best if A is in row-major, and B is in column-major order
B. best if both are in row-major order
C. best if both are in column-major order
D. independent of the storage scheme

gatecse-2004 algorithms time-complexity easy

Answer key

1.34.10 Time Complexity: GATE CSE 2004 | Question: 82

Let $A[1, \ldots, n]$ be an array storing a bit (1 or 0) at each location, and $f(m)$ is a function whose time complexity is $\Theta(m)$. Consider the following program fragment written in a C like language:

```
counter = 0;
for (i=1; i<=n; i++)
{
   if (a[i] == 1) counter++;
   else {f (counter); counter = 0;}
}
```

The complexity of this program fragment is

A. $\Omega(n^2)$
B. $\Omega(n \log n)$ and $O(n^2)$
C. $\Theta(n)$
D. $o(n)$

gatecse-2004 algorithms time-complexity normal

Answer key

1.34.11 Time Complexity: GATE CSE 2006 | Question: 15

Consider the following C-program fragment in which i, j and n are integer variables.

```
for( i = n, j = 0; i > 0; i /= 2, j +=i );
```

Let $val(j)$ denote the value stored in the variable j after termination of the for loop. Which one of the following is true?

A. $val(j) = \Theta(\log n)$
B. $val(j) = \Theta(\sqrt{n})$
C. $val(j) = \Theta(n)$
D. $val(j) = \Theta(n \log n)$

gatecse-2006 algorithms normal time-complexity

Answer key

1.34.12 Time Complexity: GATE CSE 2007 | Question: 15, ISRO2016-26

Consider the following segment of C-code:

```
int j, n;
j = 1;
while (j <= n)
   j = j * 2;
```

The number of comparisons made in the execution of the loop for any $n > 0$ is:

A. $\lceil \log_2 n \rceil + 1$
B. n
C. $\lceil \log_2 n \rceil$
D. $\lfloor \log_2 n \rfloor + 1$

gatecse-2007 algorithms time-complexity normal isro2016

Answer key

1.34.13 Time Complexity: GATE CSE 2007 | Question: 45

What is the time complexity of the following recursive function?

```
int  DoSomething (int n) {
   if (n <= 2)
      return 1;
   else
```

```
    return (DoSomething (floor (sqrt(n))) + n);
}
```

A. $\Theta(n^2)$

B. $\Theta(n \log_2 n)$

C. $\Theta(\log_2 n)$

D. $\Theta(\log_2 \log_2 n)$

gatecse-2007 algorithms time-complexity normal

Answer key

1.34.14 Time Complexity: GATE CSE 2007 | Question: 50

An array of n numbers is given, where n is an even number. The maximum as well as the minimum of these n numbers needs to be determined. Which of the following is **TRUE** about the number of comparisons needed?

A. At least $2n - c$ comparisons, for some constant c are needed.

B. At most $1.5n - 2$ comparisons are needed.

C. At least $n \log_2 n$ comparisons are needed

D. None of the above

gatecse-2007 algorithms time-complexity easy

Answer key

1.34.15 Time Complexity: GATE CSE 2007 | Question: 51

Consider the following C program segment:

```
int IsPrime (n)
  {
    int i, n;
    for (i=2; i<=sqrt(n);i++)
      if(n%i == 0)
        {printf("Not Prime \n"); return 0;}
    return 1;
  }
```

Let $T(n)$ denote number of times the for loop is executed by the program on input n. Which of the following is TRUE?

A. $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(\sqrt{n})$

B. $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(1)$

C. $T(n) = O(n)$ and $T(n) = \Omega(\sqrt{n})$

D. None of the above

gatecse-2007 algorithms time-complexity normal

Answer key

1.34.16 Time Complexity: GATE CSE 2008 | Question: 40

The minimum number of comparisons required to determine if an integer appears more than $\frac{n}{2}$ times in a sorted array of n integers is

A. $\Theta(n)$

B. $\Theta(\log n)$

C. $\Theta(\log^* n)$

D. $\Theta(1)$

gatecse-2008 normal algorithms time-complexity

Answer key

1.34.17 Time Complexity: GATE CSE 2008 | Question: 47

We have a binary heap on n elements and wish to insert n more elements (not necessarily one after another) into this heap. The total time required for this is

A. $\Theta(\log n)$

B. $\Theta(n)$

C. $\Theta(n \log n)$

D. $\Theta(n^2)$

gatecse-2008 algorithms time-complexity normal

Answer key

1.34.18 Time Complexity: GATE CSE 2008 | Question: 74

Consider the following C functions:

```
int f1 (int n)
{
    if(n == 0 || n == 1)
        return n;
    else
        return (2 * f1(n-1) + 3 * f1(n-2));
}
int f2(int n)
{
    int i;
    int X[N], Y[N], Z[N];
    X[0] = Y[0] = Z[0] = 0;
    X[1] = 1; Y[1] = 2; Z[1] = 3;
    for(i = 2; i <= n; i++){
        X[i] = Y[i-1] + Z[i-2];
        Y[i] = 2 * X[i];
        Z[i] = 3 * X[i];
    }
    return X[n];
}
```

The running time of $f1(n)$ and $f2(n)$ are

A. $\Theta(n)$ and $\Theta(n)$
B. $\Theta(2^n)$ and $\Theta(n)$
C. $\Theta(n)$ and $\Theta(2^n)$
D. $\Theta(2^n)$ and $\Theta(2^n)$

gatecse-2008 algorithms time-complexity normal

Answer key

1.34.19 Time Complexity: GATE CSE 2008 | Question: 75

Consider the following C functions:

```
int f1 (int n)
{
    if(n == 0 || n == 1)
        return n;
    else
        return (2 * f1(n-1) + 3 * f1(n-2));
}
int f2(int n)
{
    int i;
    int X[N], Y[N], Z[N];
    X[0] = Y[0] = Z[0] = 0;
    X[1] = 1; Y[1] = 2; Z[1] = 3;
    for(i = 2; i <= n; i++){
        X[i] = Y[i-1] + Z[i-2];
        Y[i] = 2 * X[i];
        Z[i] = 3 * X[i];
    }
    return X[n];
}
```

$f1(8)$ and $f2(8)$ return the values

A. 1661 and 1640
B. 59 and 59
C. 1640 and 1640
D. 1640 and 1661

gatecse-2008 normal algorithms time-complexity

Answer key

1.34.20 Time Complexity: GATE CSE 2010 | Question: 12

Two alternative packages A and B are available for processing a database having 10^k records. Package A requires $0.0001n^2$ time units and package B requires $10n\log_{10} n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

A. 12
B. 10
C. 6
D. 5

gatecse-2010 algorithms time-complexity easy

Answer key

1.34.21 Time Complexity: GATE CSE 2014 Set 1 | Question: 42

Consider the following pseudo code. What is the total number of multiplications to be performed?

```
D = 2
for i = 1 to n do
   for j = i to n do
      for k = j + 1 to n do
         D = D * 3
```

A. Half of the product of the 3 consecutive integers.
B. One-third of the product of the 3 consecutive integers.
C. One-sixth of the product of the 3 consecutive integers.
D. None of the above.

gatecse-2014-set1 algorithms time-complexity normal

Answer key

1.34.22 Time Complexity: GATE CSE 2015 Set 1 | Question: 40

An algorithm performs $(\log N)^{\frac{1}{2}}$ find operations , N insert operations, $(\log N)^{\frac{1}{2}}$ delete operations, and $(\log N)^{\frac{1}{2}}$ decrease-key operations on a set of data items with keys drawn from a linearly ordered set . For a delete operation, a pointer is provided to the record that must be deleted . For the decrease-key operation, a pointer is provided to the record that has its key decreased. Which one of the following data structures is the most suited for the algorithm to use, if the goal is to achieve the best total asymptotic complexity considering all the operations?

A. Unsorted array
B. Min - heap
C. Sorted array
D. Sorted doubly linked list

gatecse-2015-set1 algorithms data-structures normal time-complexity

Answer key

1.34.23 Time Complexity: GATE CSE 2015 Set 2 | Question: 22

An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is

A. $\Theta(n \log n)$
B. $\Theta(n)$
C. $\Theta(\log n)$
D. $\Theta(1)$

gatecse-2015-set2 algorithms time-complexity easy

Answer key

1.34.24 Time Complexity: GATE CSE 2017 Set 2 | Question: 03

Match the algorithms with their time complexities:

Algorithms	**Time Complexity**
P. Tower of Hanoi with n disks	i. $\Theta(n^2)$
Q. Binary Search given n sorted numbers	ii. $\Theta(n \log n)$
R. Heap sort given n numbers at the worst case	iii. $\Theta(2^n)$
S. Addition of two $n \times n$ matrices	iv. $\Theta(\log n)$

A. $P \to (iii) \quad Q \to (iv) \quad r \to (i) \quad S \to (ii)$
B. $P \to (iv) \quad Q \to (iii) \quad r \to (i) \quad S \to (ii)$
C. $P \to (iii) \quad Q \to (iv) \quad r \to (ii) \quad S \to (i)$
D. $P \to (iv) \quad Q \to (iii) \quad r \to (ii) \quad S \to (i)$

gatecse-2017-set2 algorithms time-complexity match-the-following easy

Answer key

1.34.25 Time Complexity: GATE CSE 2017 Set 2 | Question: 38

Consider the following C function

```
int fun(int n) {
    int i, j;
    for(i=1; i<=n; i++) {
        for (j=1; j<n; j+=i) {
            printf("%d %d", i, j);
        }
    }
}
```

Time complexity of fun in terms of Θ notation is

A. $\Theta(n\sqrt{n})$
B. $\Theta(n^2)$
C. $\Theta(n \log n)$
D. $\Theta(n^2 \log n)$

gatecse-2017-set2 algorithms time-complexity

Answer key

1.34.26 Time Complexity: GATE CSE 2019 | Question: 37

There are n unsorted arrays: $A_1, A_2, \dots, A_n$. Assume that n is odd.Each of $A_1, A_2, \dots, A_n$ contains n distinct elements. There are no common elements between any two arrays. The worst-case time complexity of computing the median of the medians of $A_1, A_2, \dots, A_n$ is

A. $O(n)$
B. $O(n \log n)$
C. $O(n^2)$
D. $\Omega(n^2 \log n)$

gatecse-2019 algorithms time-complexity two-marks

Answer key

1.34.27 Time Complexity: GATE CSE 2024 | Set 1 | Question: 7

Given an integer array of size N, we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

A. both $\mathrm{O}(N)$ and $\Omega(N)$
B. $\mathrm{O}(N)$ but $\text{not}\, \Omega(N)$
C. $\Omega(N)$ but not $\mathrm{O}(N)$
D. neither $\mathrm{O}(N)$ nor $\Omega(N)$

gatecse2024-set1 algorithms time-complexity one-mark

Answer key

1.34.28 Time Complexity: GATE IT 2007 | Question: 17

Exponentiation is a heavily used operation in public key cryptography. Which of the following options is the tightest upper bound on the number of multiplications required to compute $b^n \bmod m, 0 \leq b, n \leq m$?

A. $O(\log n)$
B. $O(\sqrt{n})$
C. $O\left(\frac{n}{\log n}\right)$
D. $O(n)$

gateit-2007 algorithms time-complexity normal

Answer key

1.34.29 Time Complexity: GATE IT 2007 | Question: 81

Let $P_1, P_2, \dots, P_n$ be n points in the xy-plane such that no three of them are collinear. For every pair of points P_i and P_j, let L_{ij} be the line passing through them. Let L_{ab} be the line with the steepest gradient among all $n(n-1)/2$ lines.

The time complexity of the best algorithm for finding P_a and P_b is

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n \log^2 n)$
D. $\Theta(n^2)$

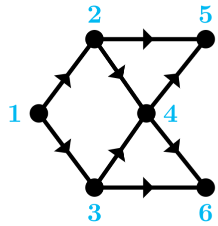
gateit-2007 algorithms time-complexity normal

Answer key

1.35 Topological Sort (4)

1.35.1 Topological Sort: GATE CSE 2007 | Question: 5

Consider the DAG with $V = \{1, 2, 3, 4, 5, 6\}$ shown below.



Which of the following is not a topological ordering?

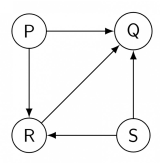
A. $1\,2\,3\,4\,5\,6$ B. $1\,3\,2\,4\,5\,6$ C. $1\,3\,2\,4\,6\,5$ D. $3\,2\,4\,1\,6\,5$

gatecse-2007 algorithms graph-algorithms topological-sort easy

Answer key

1.35.2 Topological Sort: GATE CSE 2014 Set 1 | Question: 13

Consider the directed graph below given.



Which one of the following is **TRUE**?

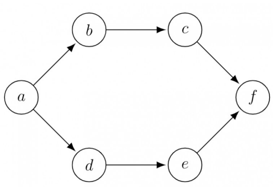
A. The graph does not have any topological ordering.
B. Both PQRS and SRQP are topological orderings.
C. Both PSRQ and SPRQ are topological orderings.
D. PSRQ is the only topological ordering.

gatecse-2014-set1 graph-algorithms easy topological-sort

Answer key

1.35.3 Topological Sort: GATE CSE 2016 Set 1 | Question: 11

Consider the following directed graph:



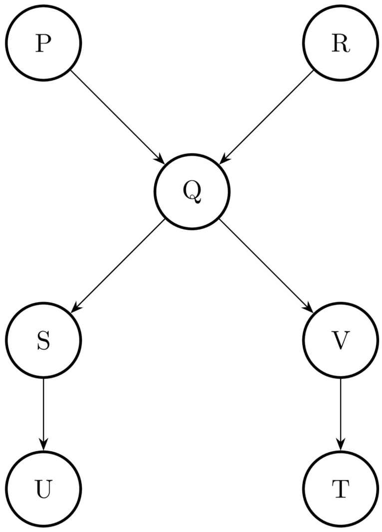
The number of different topological orderings of the vertices of the graph is ______________.

gatecse-2016-set1 algorithms graph-algorithms normal numerical-answers topological-sort

Answer key

1.35.4 Topological Sort: GATE DS&AI 2024 | Question: 41

Consider the directed acyclic graph (DAG) below:



Which of the following is/are valid vertex orderings that can be obtained from a topological sort of the DAG?

A. P Q R S T U V
B. P R Q V S U T
C. P Q R S V U T
D. P R Q S V T U

gate-ds-ai-2024 algorithms topological-sort directed-acyclic-graph multiple-selects two-marks

Answer key☟

Answer Keys

1.0.1	6:6	1.1.1	N/A	1.1.2	N/A	1.1.3	B	1.1.4	C
1.1.5	12	1.1.6	29	1.1.7	A;C	1.1.8	B	1.2.1	N/A
1.2.2	N/A	1.2.3	C	1.2.4	A	1.2.5	B	1.2.6	A
1.2.7	C	1.2.8	C	1.2.9	C	1.3.1	X	1.3.2	B
1.3.3	D	1.3.4	A	1.3.5	A;C	1.3.6	A;D	1.3.7	A
1.4.1	A;B	1.4.2	B	1.4.3	D	1.4.4	B	1.4.5	A
1.4.6	C	1.4.7	C	1.4.8	D	1.4.9	A	1.4.10	C
1.4.11	C	1.4.12	D	1.4.13	D	1.4.14	A	1.5.1	B
1.5.2	C	1.6.1	B	1.6.2	C	1.6.3	A	1.7.1	D
1.8.1	C	1.9.1	N/A	1.9.2	B	1.9.3	D	1.9.4	A
1.9.5	D	1.9.6	D	1.10.1	75:75	1.11.1	10:10	1.12.1	B
1.12.2	C	1.12.3	C	1.12.4	B	1.12.5	B	1.12.6	A
1.12.7	34	1.12.8	150	1.12.9	C	1.13.1	B	1.13.2	D
1.13.3	A	1.13.4	A	1.13.5	B	1.13.6	A	1.13.7	A;B
1.13.8	929 : 929	1.13.9	A	1.13.10	C	1.13.11	D	1.14.1	N/A
1.14.2	C	1.14.3	C	1.14.4	C	1.14.5	D	1.14.6	D
1.14.7	C	1.14.8	C	1.14.9	B	1.14.10	19	1.14.11	D
1.14.12	31	1.14.13	D	1.14.14	A	1.14.15	5040	1.14.16	C
1.14.17	60	1.14.18	A	1.14.19	D	1.14.20	D	1.14.21	D
1.14.22	B	1.15.1	A	1.15.2	B	1.15.3	D	1.15.4	A
1.15.5	16	1.16.1	B	1.16.2	N/A	1.16.3	13	1.16.4	C

1.16.5	B	1.16.6	C	1.16.7	D	1.17.1	2.33	1.17.2	A
1.17.3	D	1.17.4	225	1.17.5	B	1.17.6	A	1.18.1	N/A
1.18.2	N/A	1.18.3	C	1.18.4	A	1.18.5	N/A	1.18.6	C
1.18.7	C	1.18.8	N/A	1.18.9	C	1.18.10	D	1.18.11	A
1.18.12	B	1.18.13	D	1.18.14	C	1.18.15	C	1.18.16	D
1.18.17	D	1.18.18	D	1.18.19	B	1.18.20	C	1.18.21	B
1.18.22	D	1.18.23	B	1.18.24	A	1.18.25	9	1.18.26	A
1.18.27	D	1.18.28	51	1.18.29	0	1.18.30	D	1.18.31	81
1.18.32	1023 : 1023	1.18.33	15 : 15	1.18.34	D	1.18.35	C	1.18.36	C
1.18.37	D	1.18.38	C	1.19.1	A	1.19.2	D	1.19.3	A;C
1.20.1	D	1.21.1	C	1.21.2	1500	1.21.3	C	1.22.1	B
1.22.2	B	1.22.3	B	1.23.1	C	1.23.2	358	1.24.1	B;D;E
1.24.2	N/A	1.24.3	2	1.24.4	N/A	1.24.5	N/A	1.24.6	C
1.24.7	N/A	1.24.8	B	1.24.9	C	1.24.10	B	1.24.11	D
1.24.12	D	1.24.13	D	1.24.14	D	1.24.15	B	1.24.16	B
1.24.17	C	1.24.18	X	1.24.19	6	1.24.20	69	1.24.21	995
1.24.22	A	1.24.23	7	1.24.24	B	1.24.25	4	1.24.26	D
1.24.27	99	1.24.28	3 : 3	1.24.29	C	1.24.30	A;B;C	1.24.31	24
1.24.32	9	1.24.33	5:5	1.24.34	A	1.25.1	D	1.25.2	C
1.26.1	C	1.26.2	N/A	1.26.3	N/A	1.26.4	C	1.26.5	C
1.26.6	B	1.26.7	B	1.26.8	B	1.26.9	C	1.26.10	A
1.26.11	B	1.26.12	A	1.26.13	0.08	1.26.14	0	1.27.1	N/A
1.27.2	N/A	1.27.3	N/A	1.27.4	N/A	1.27.5	N/A	1.27.6	N/A
1.27.7	N/A	1.27.8	B	1.27.9	A	1.27.10	N/A	1.27.11	B
1.27.12	N/A	1.27.13	B	1.27.14	C	1.27.15	B	1.27.16	D
1.27.17	A	1.27.18	B	1.27.19	D	1.27.20	X	1.27.21	A
1.27.22	D	1.27.23	A	1.27.24	A	1.27.25	B	1.27.26	2.32 : 2.33
1.27.27	B	1.27.28	A	1.27.29	C	1.27.30	C	1.27.31	A
1.27.32	A	1.27.33	B	1.27.34	C	1.27.35	A	1.28.1	C
1.28.2	A	1.28.3	10230	1.28.4	60 : 60	1.29.1	N/A	1.29.2	C
1.29.3	A	1.29.4	C	1.29.5	A	1.29.6	5	1.29.7	B;C;D
1.30.1	N/A	1.30.2	B	1.30.3	D	1.30.4	A	1.30.5	C
1.30.6	B	1.30.7	C	1.30.8	A;C;D	1.30.9	C	1.31.1	N/A
1.31.2	A	1.31.3	7	1.31.4	N/A	1.31.5	D	1.31.6	N/A
1.31.7	N/A	1.31.8	B	1.31.9	B	1.31.10	N/A	1.31.11	N/A
1.31.12	B	1.31.13	A	1.31.14	C	1.31.15	A	1.31.16	A
1.31.17	C	1.31.18	B	1.31.19	147.1 : 148.1	1.31.20	D	1.31.21	D
1.31.22	C	1.31.23	D	1.31.24	3	1.31.25	A	1.31.26	B
1.31.27	C	1.31.28	C	1.32.1	B	1.33.1	C	1.33.2	109

1.33.3	B
1.34.5	A
1.34.10	C
1.34.15	B
1.34.20	C
1.34.25	C
1.35.1	D

1.34.1	N/A
1.34.6	N/A
1.34.11	C
1.34.16	B
1.34.21	C
1.34.26	C
1.35.2	C

1.34.2	N/A
1.34.7	A
1.34.12	D
1.34.17	B
1.34.22	A
1.34.27	A
1.35.3	6

1.34.3	B;C;D;E
1.34.8	C
1.34.13	D
1.34.18	B
1.34.23	D
1.34.28	A
1.35.4	B;D

1.34.4	D
1.34.9	D
1.34.14	B
1.34.19	C
1.34.24	C
1.34.29	B